POUMONS



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Un effort sain…

dans une vie qui respire



Plan du cours

- Généralités
- Configuration externe
- Plèvres
- Rapports

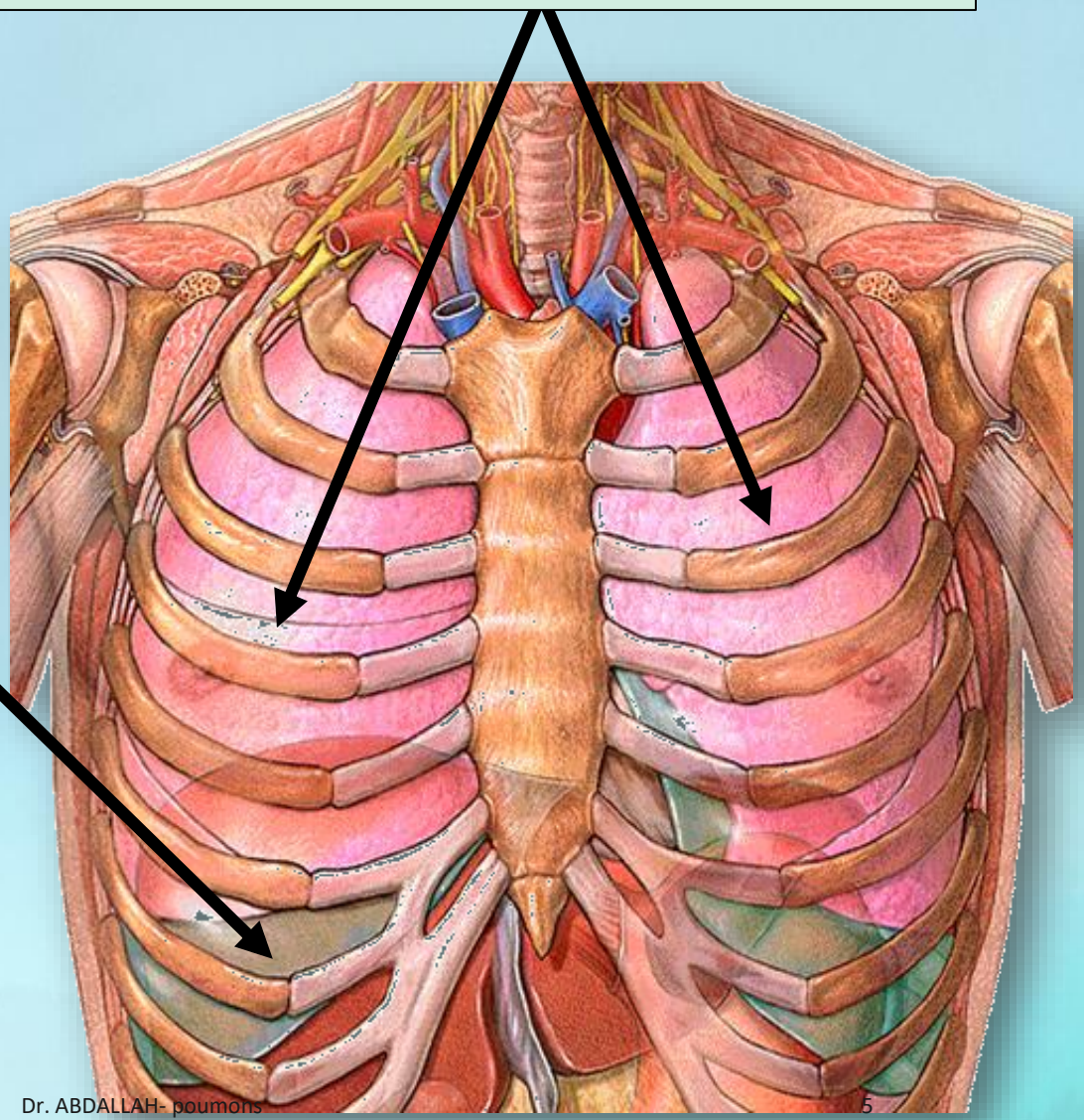


Généralités

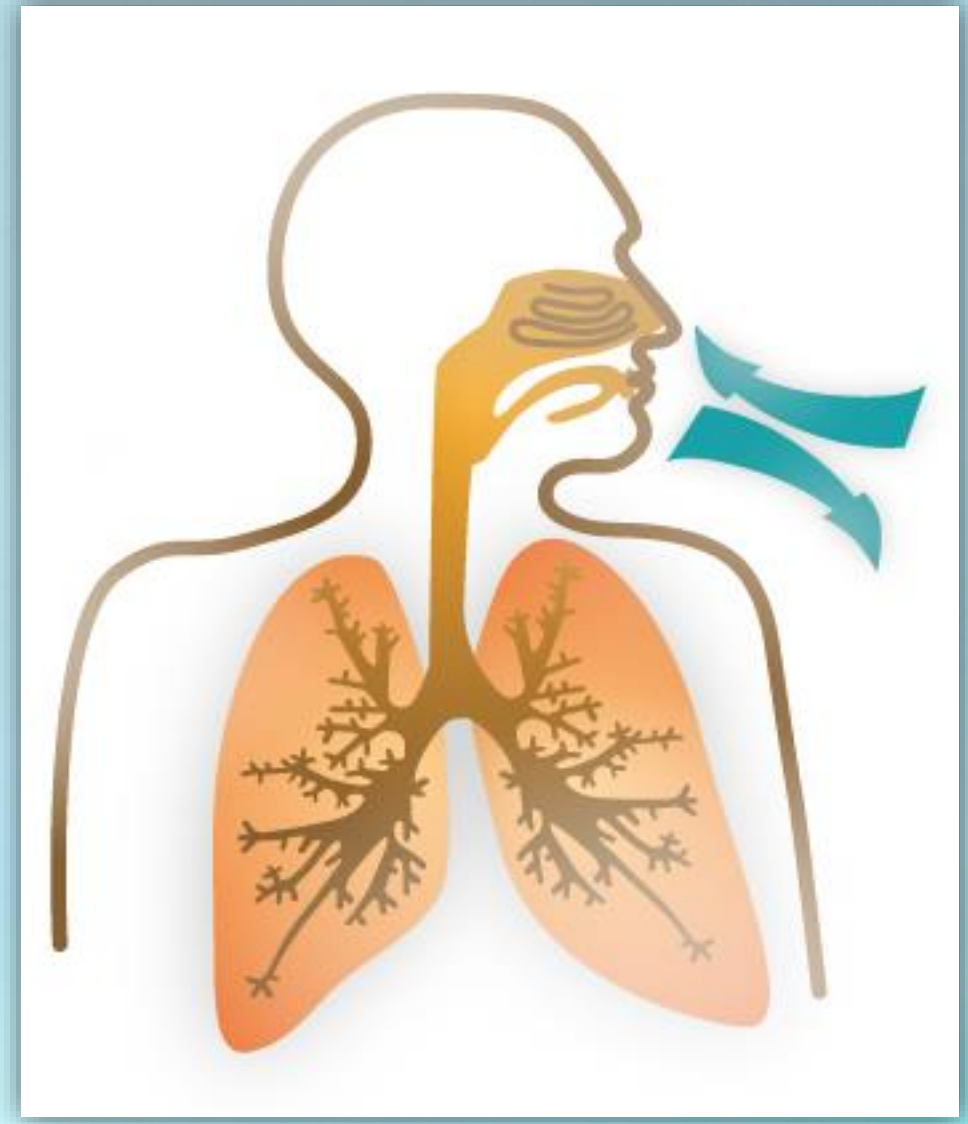


Poumons ou **organes de la respiration**, sont:

- **2 organes intra-thoraciques**
- Enveloppés par: **plèvres**
- Assurent **3 rôles**…



1 Assurent:
hématose
(échanges gazeux)



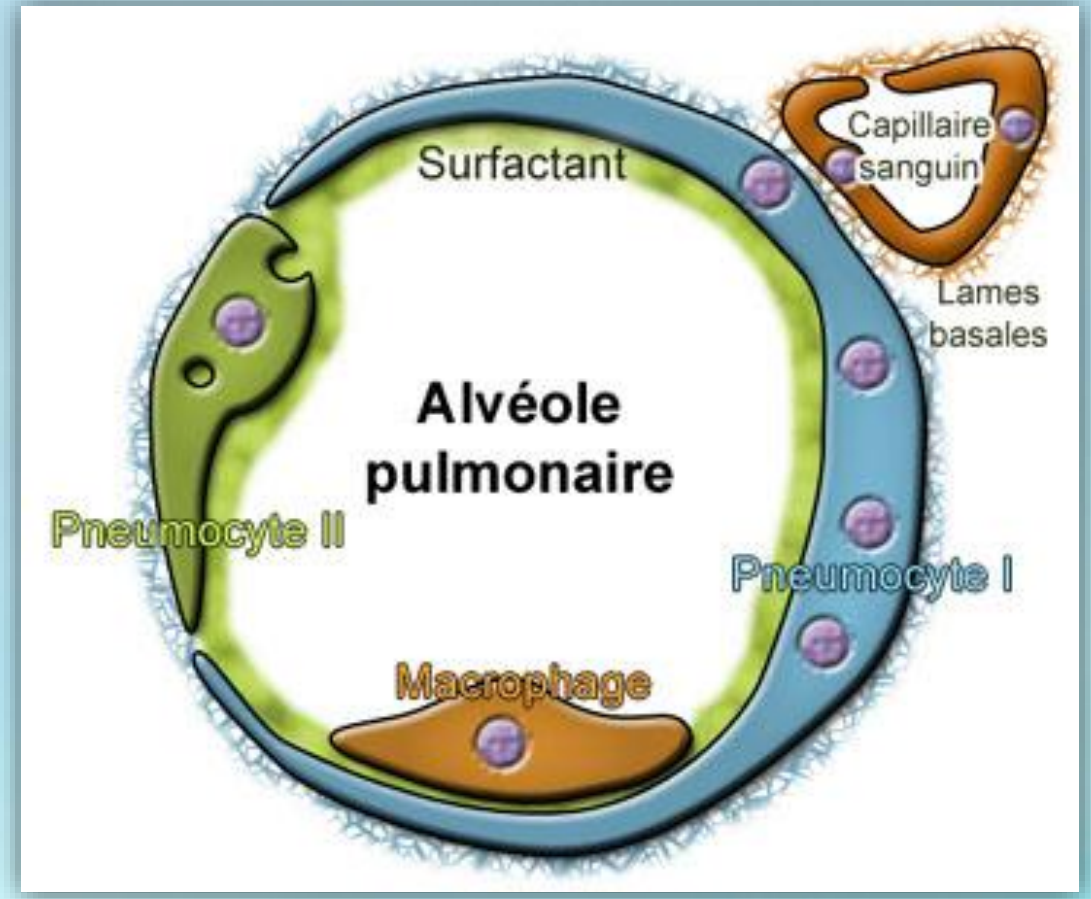
2 **Protègent** contre: **agents pathogènes inhalés**





3

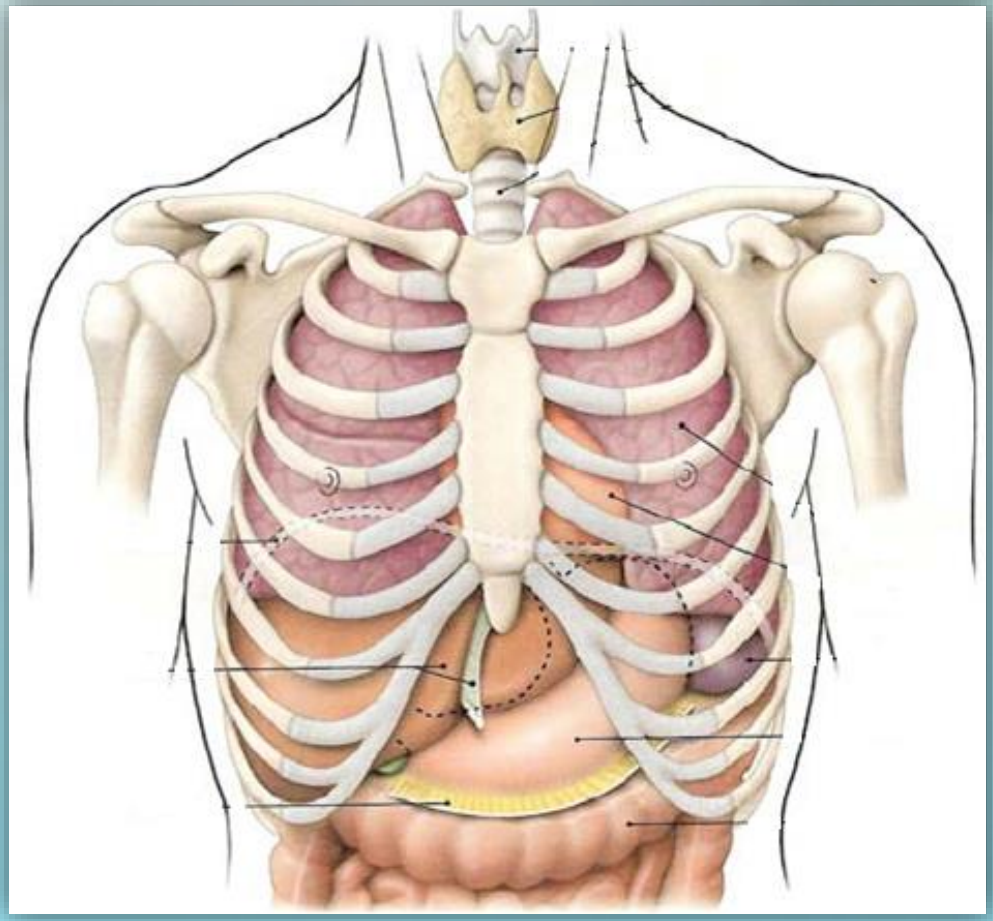
Produisent:
Surfactant
(agent tensio-actif de la stabilité alvéolaire)



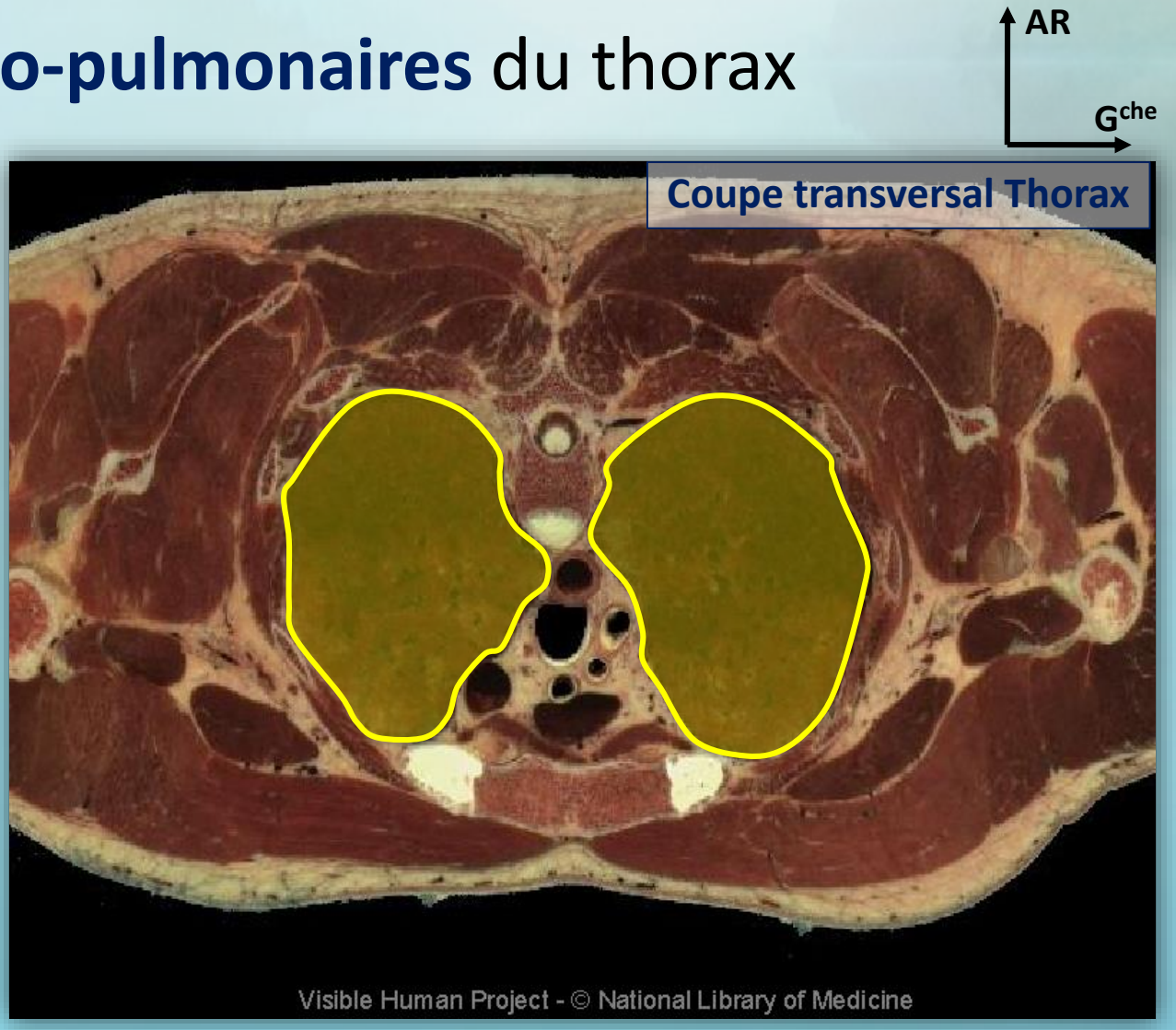
Situation

- Dans:

cage thoracique

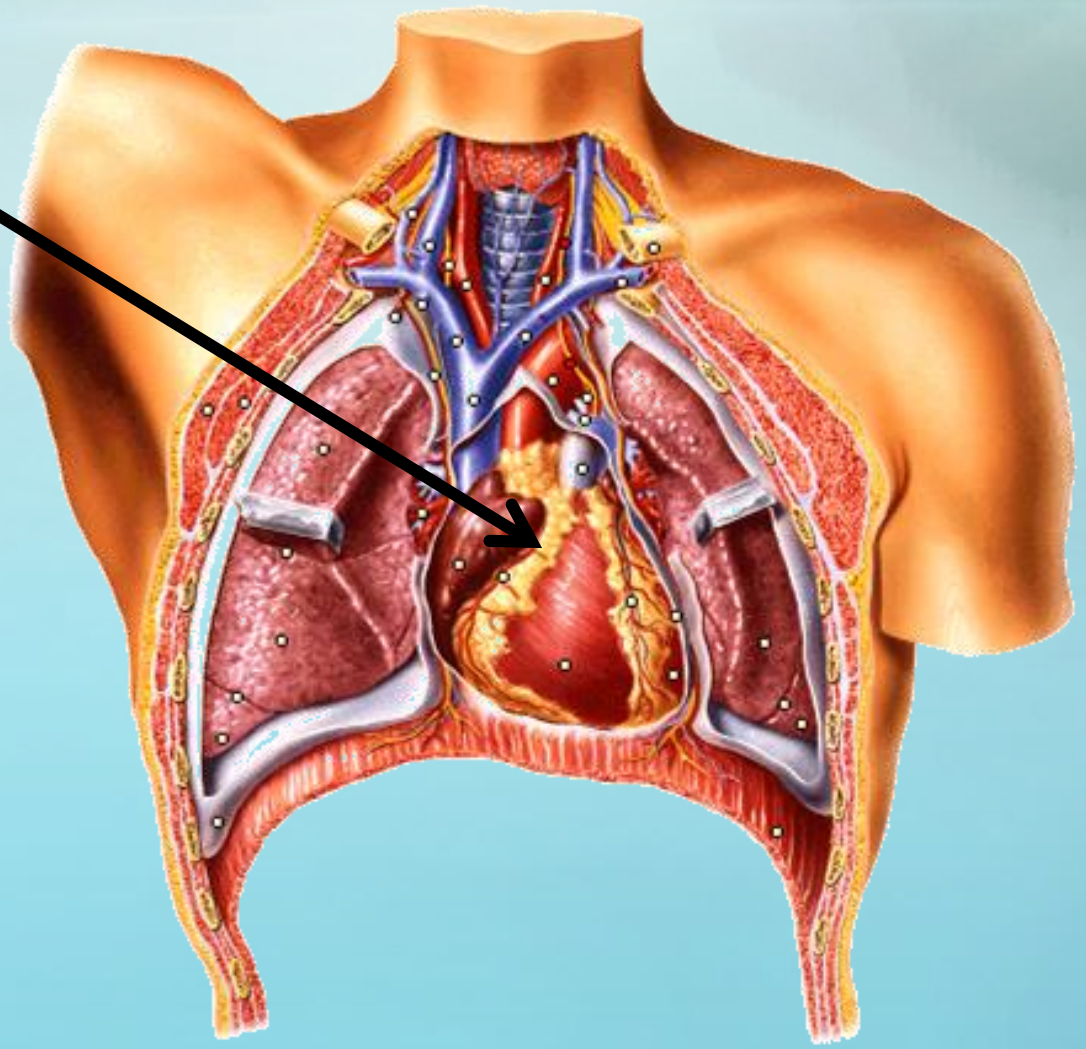


- Occupant:
  cavités pleuro-pulmonaires du thorax



- De part et d’autre du:

médiastin



Surface & Consistance

- **Surface lisse et brillante**
- **Consistance spongieuse et élastique**

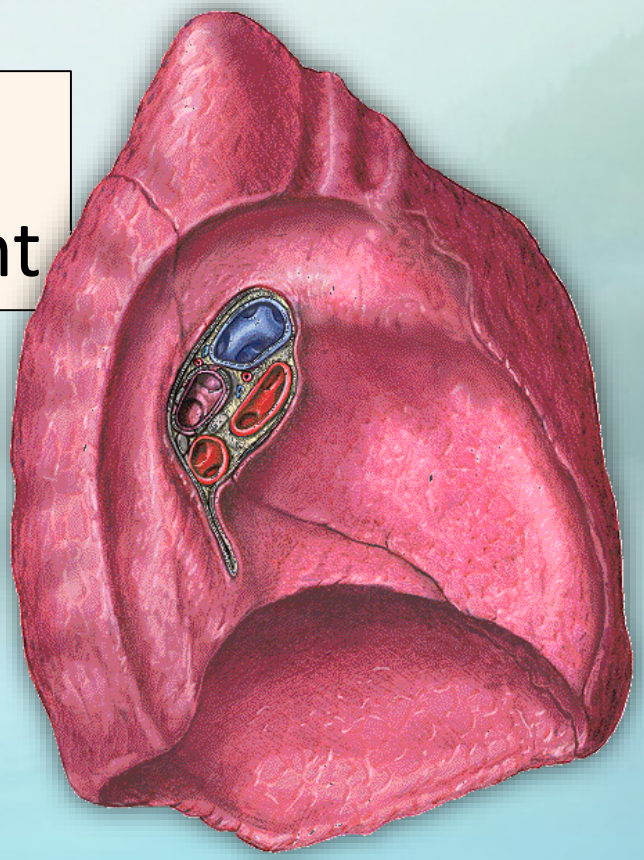




Coloration

Rose claire:

chez nourrisson et enfant





Avec **traits** et **tâches noirâtres:**

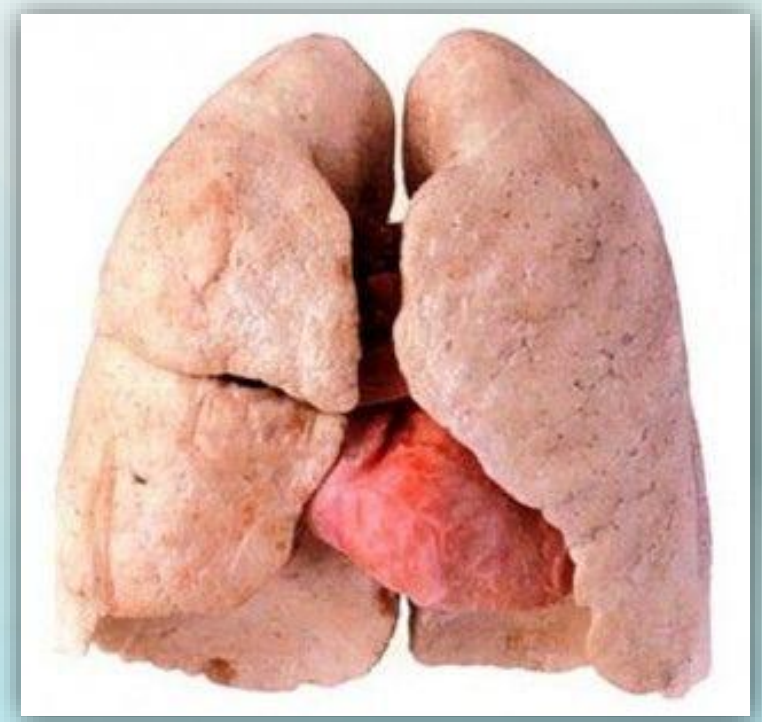
chez adulte

Remarque:

Environnement pollué

(cigarettes, fumée, poussières..)

Tâches plus prononcées





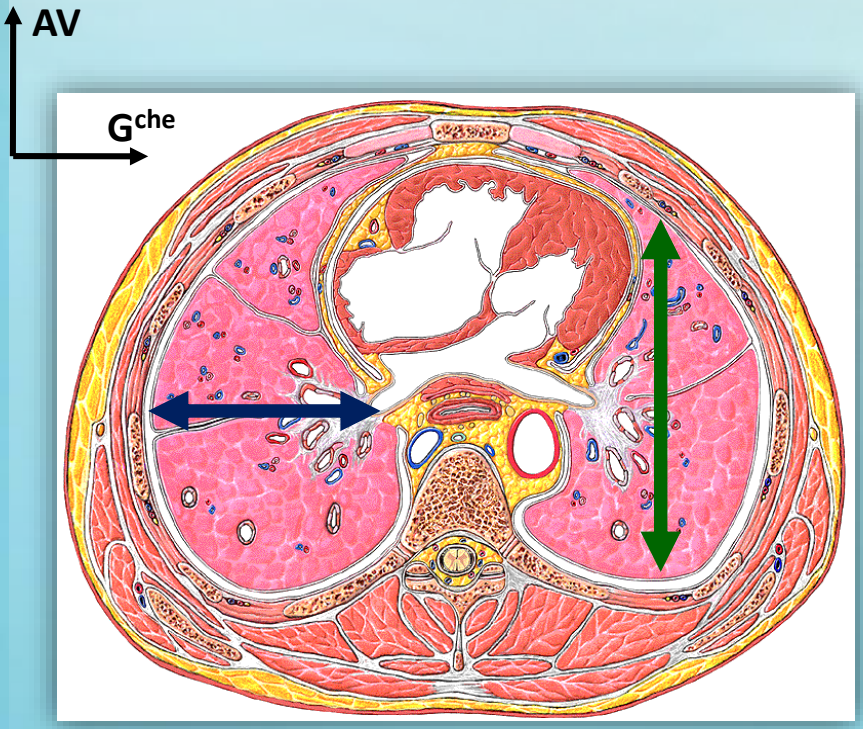
Mensurations

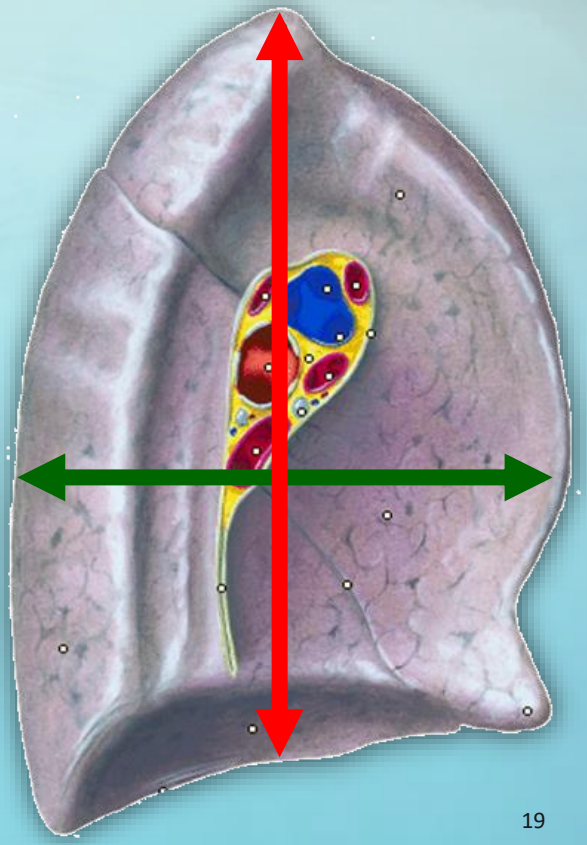
Le poumon droit est **plus volumineux** que **le gauche**

Largeur:
7 à 10 cm

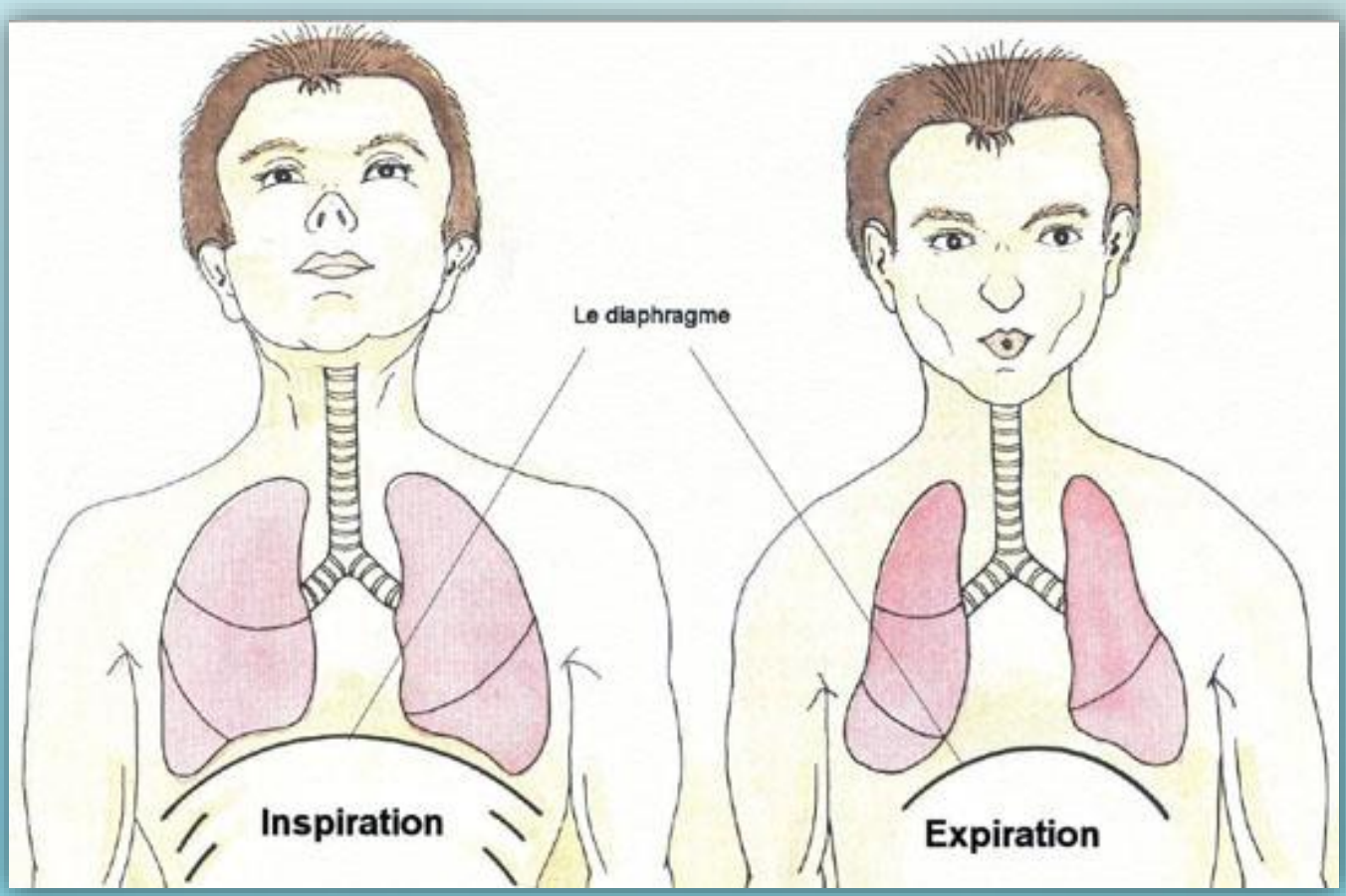
Épaisseur:
16 cm

Hauteur:
25 cm



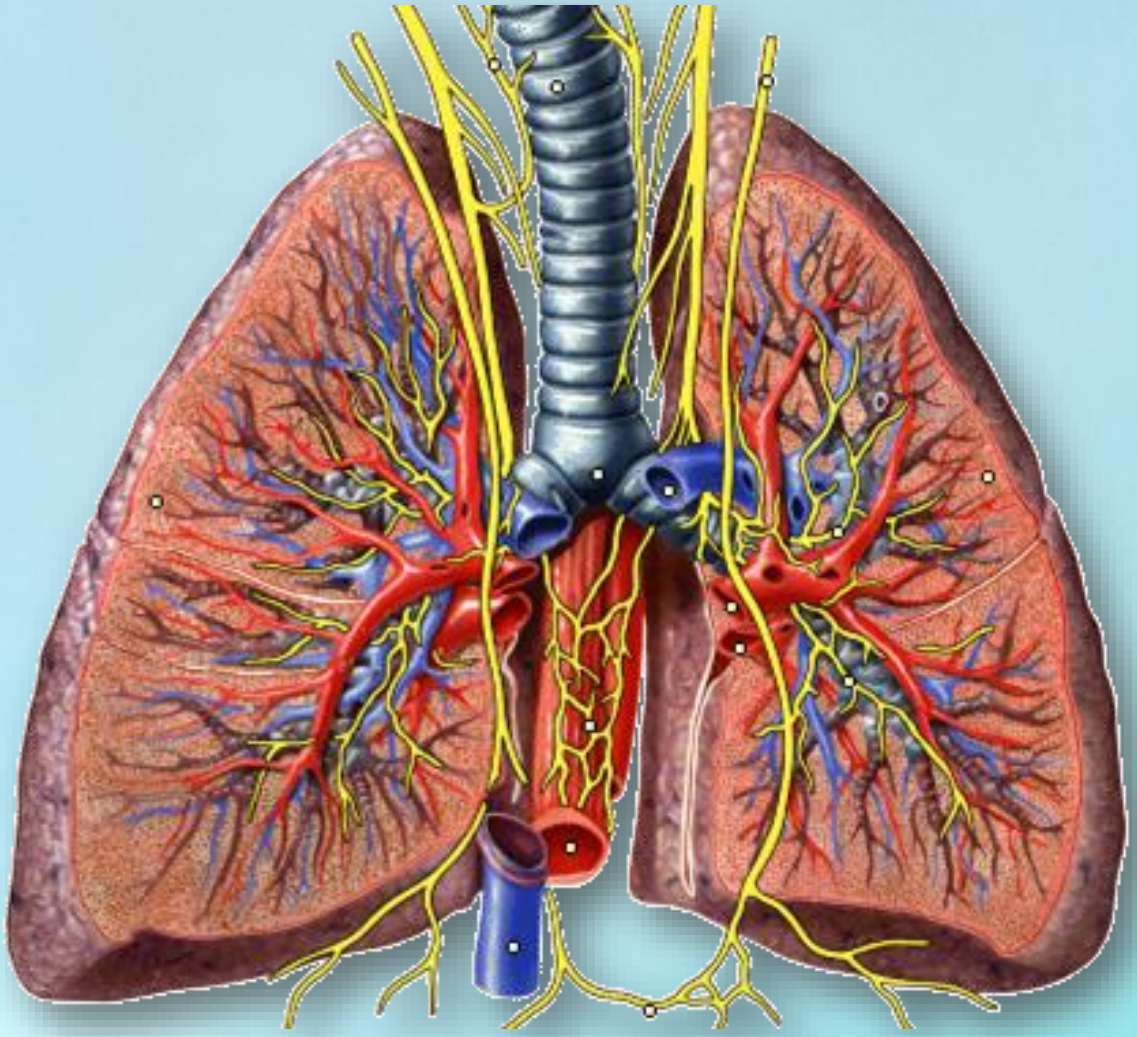


- **Poids**: **550 g** (gauche) à **650 g** (droit)
- **Capacité respiratoire moyenne** = **5 litres**



Structure

- **Parenchyme pulmonaire** hétérogène
- Formé de: **4 éléments**…

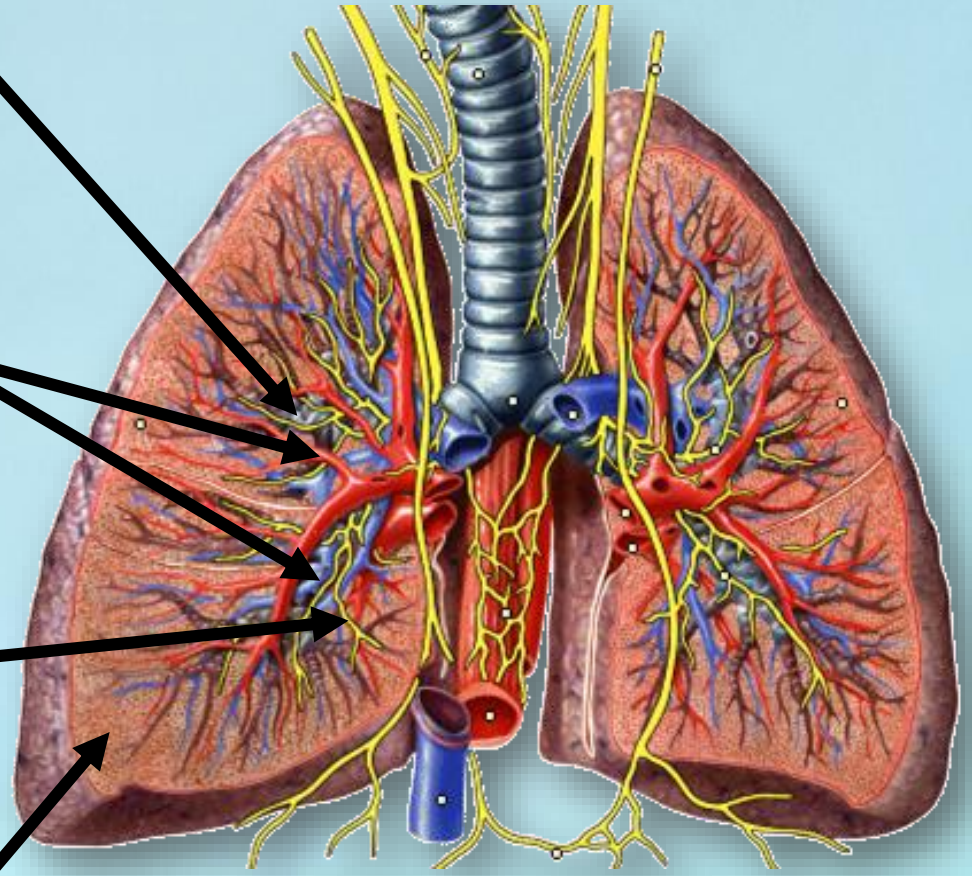


(1)
Arbre bronchique
(partie intrapulmonaire)

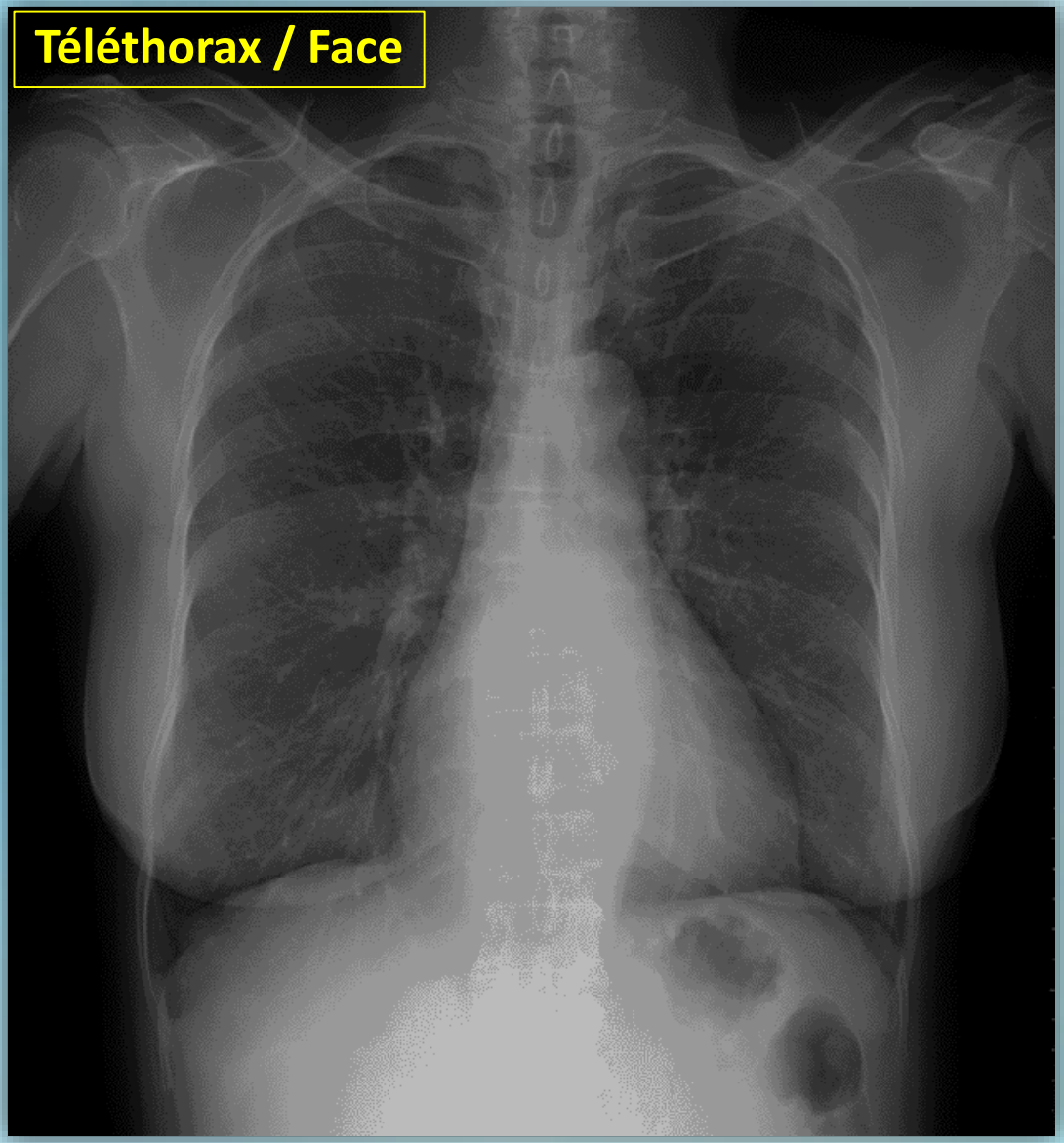
(2)
Vaisseaux
pulmonaires et **bronchiques**
et **les lymphatiques**

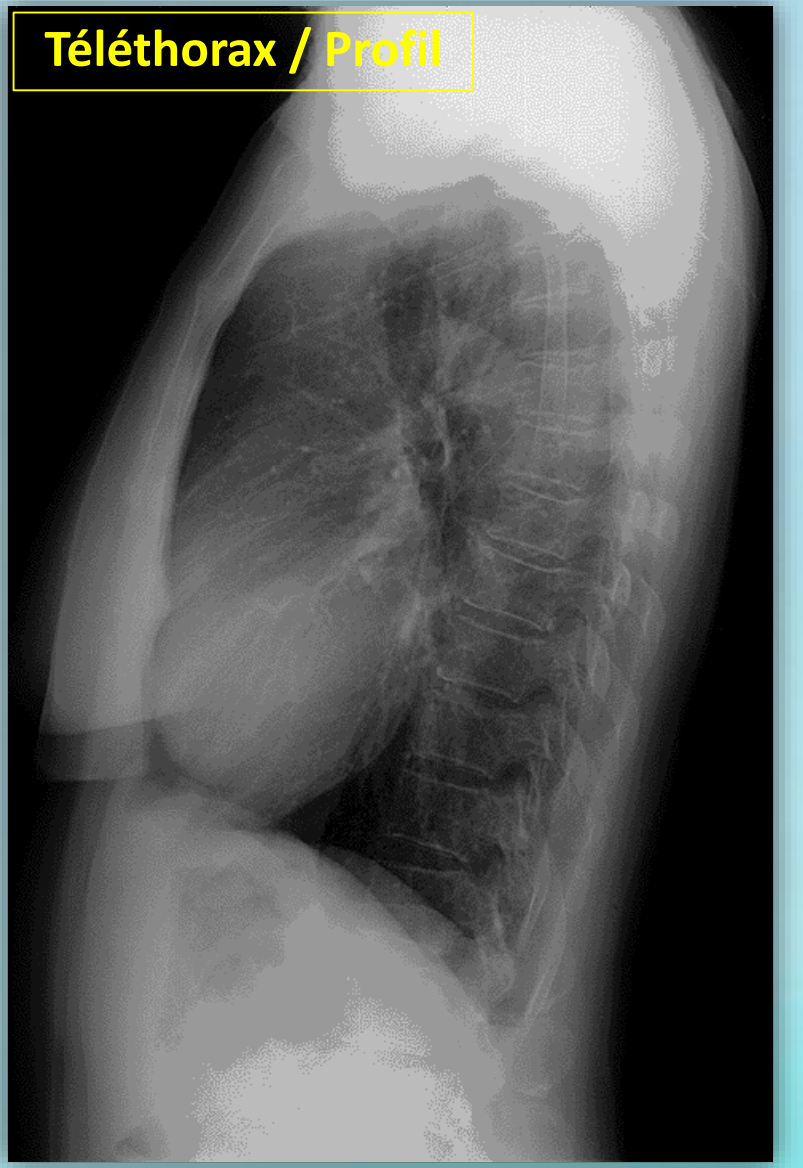
(3)
Rameaux nerveux

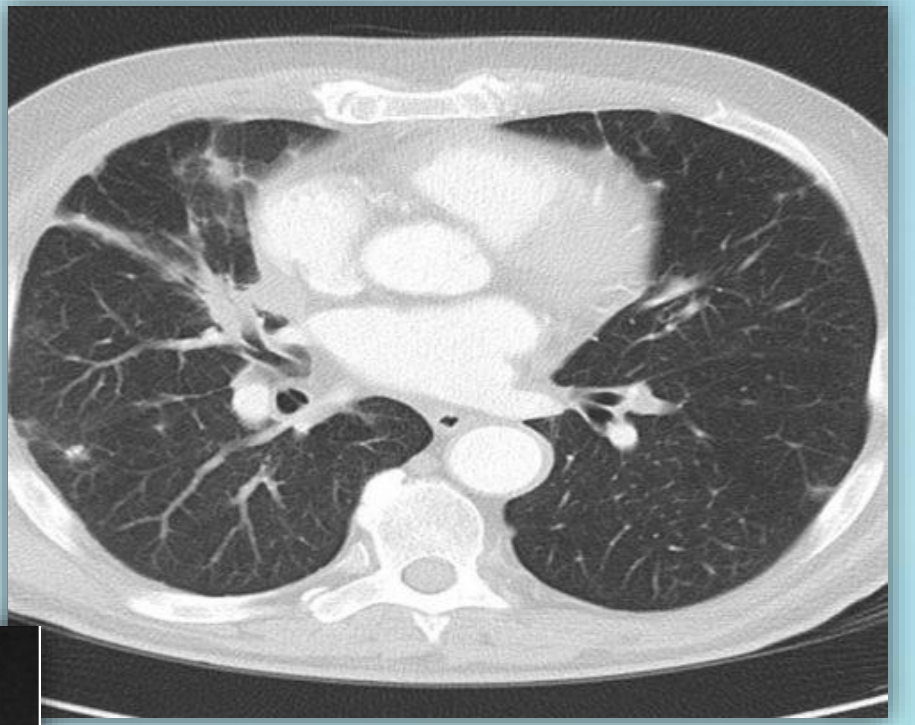
(4)
Tissu conjonctif
(de remplissage)



Exploration

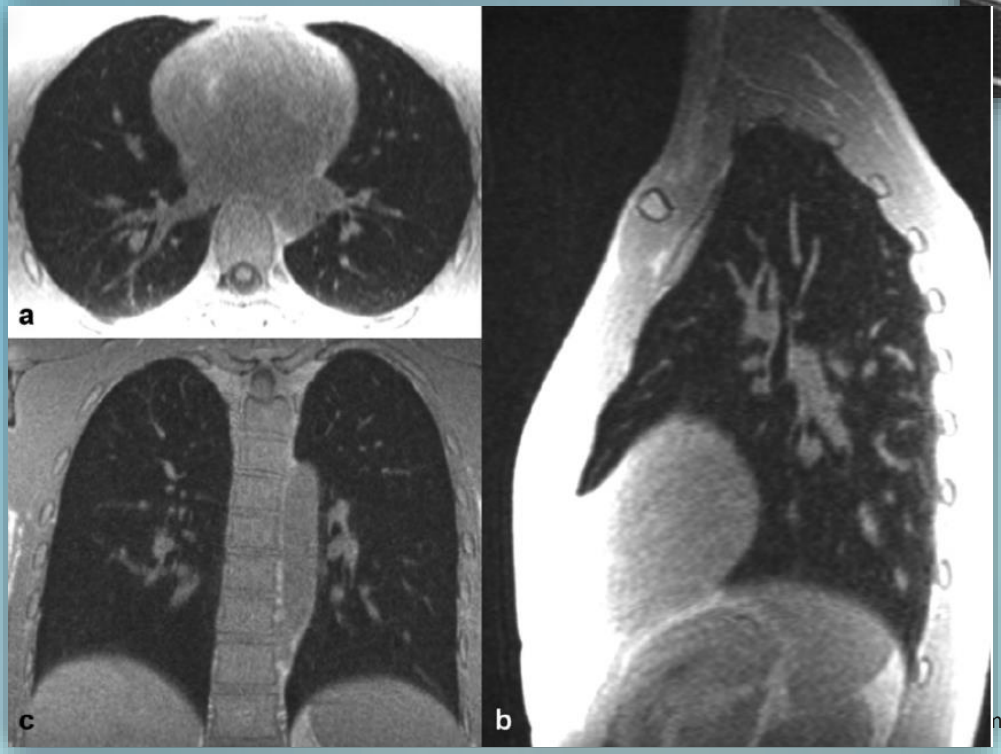

Téléthorax / Face

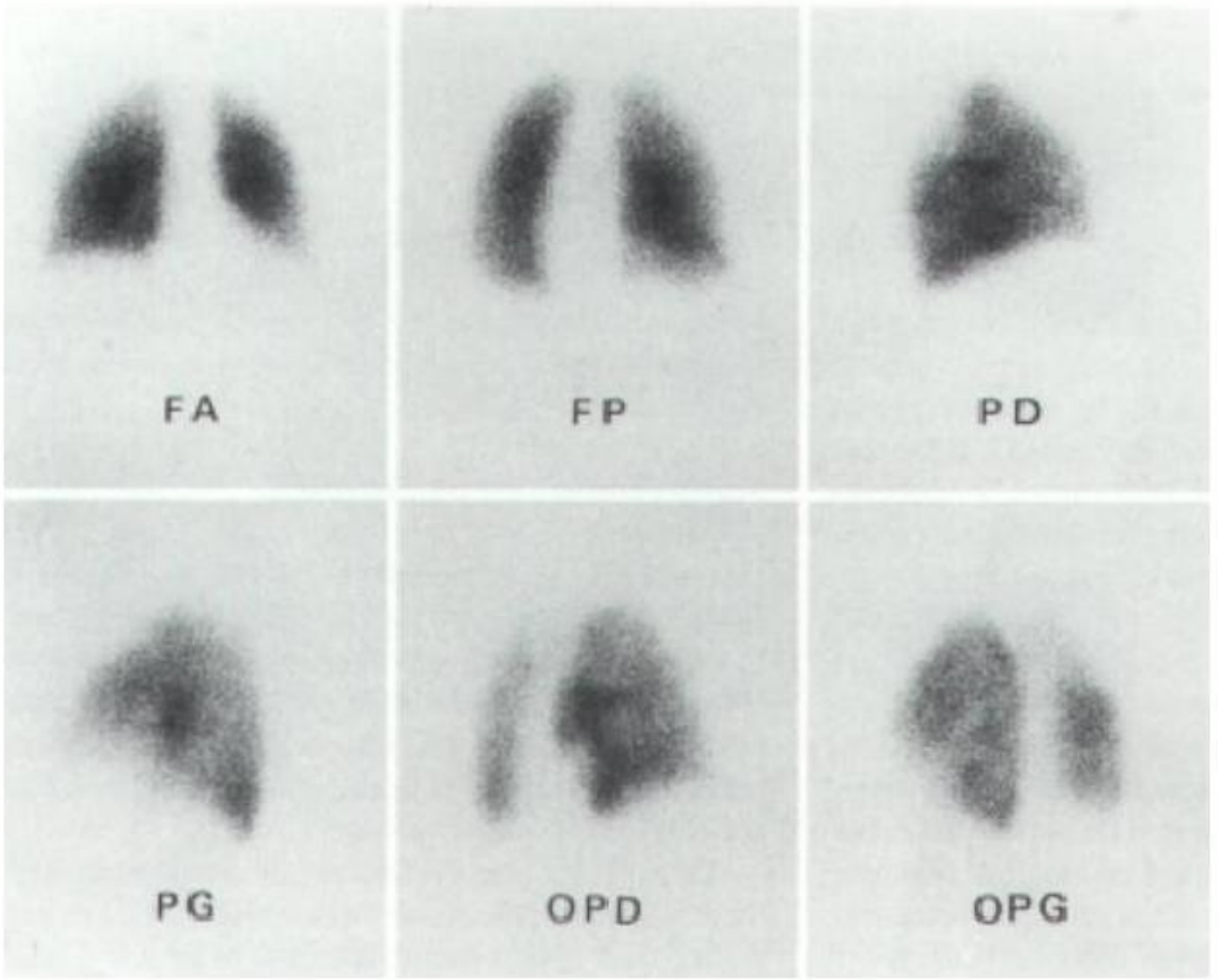

Téléthorax / Profil



TDM

IRM





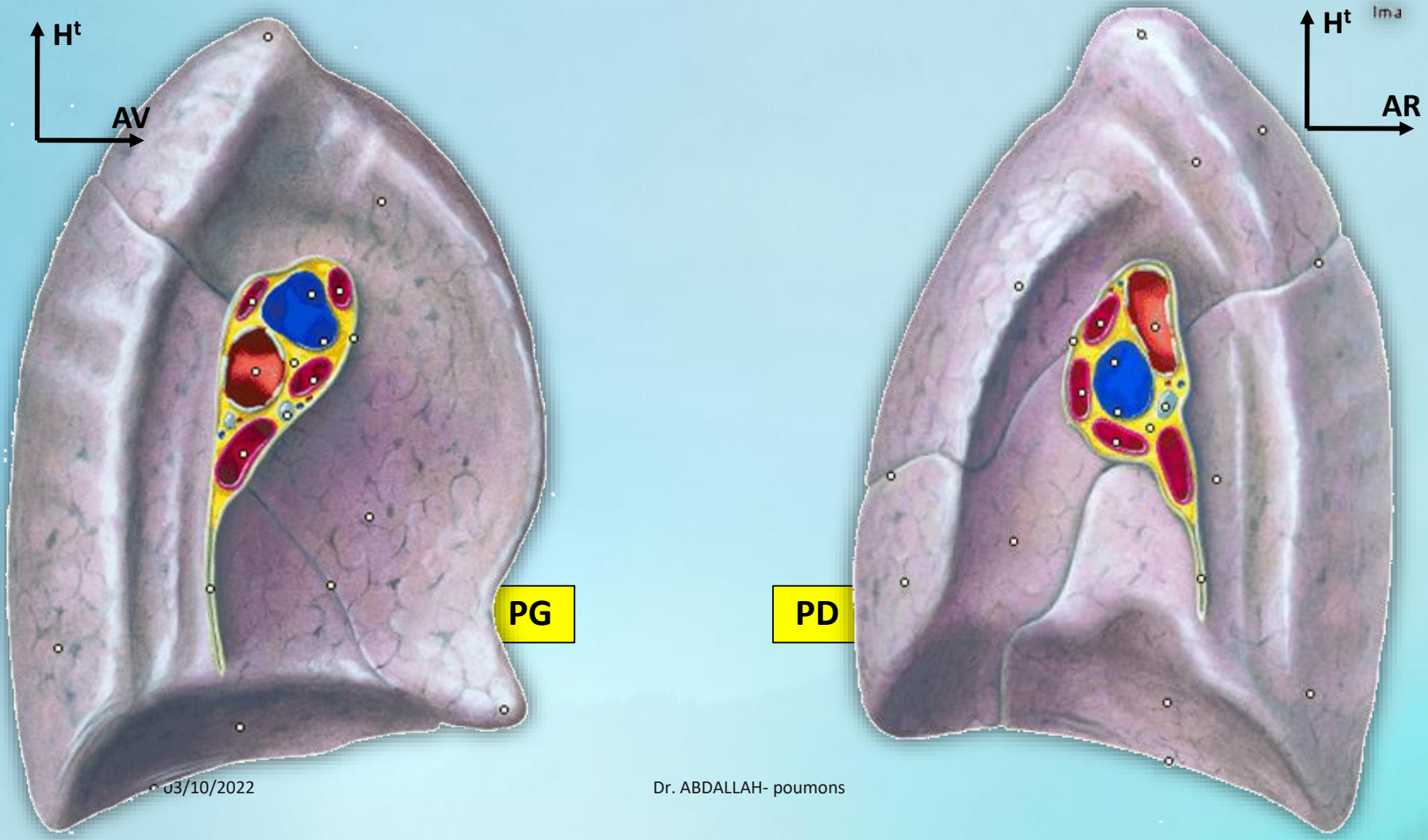
SCINTIGRAPHIE DE VENTILATION

Configuration externe



Poumon ressemble à:

une **pyramide demi-cônique**



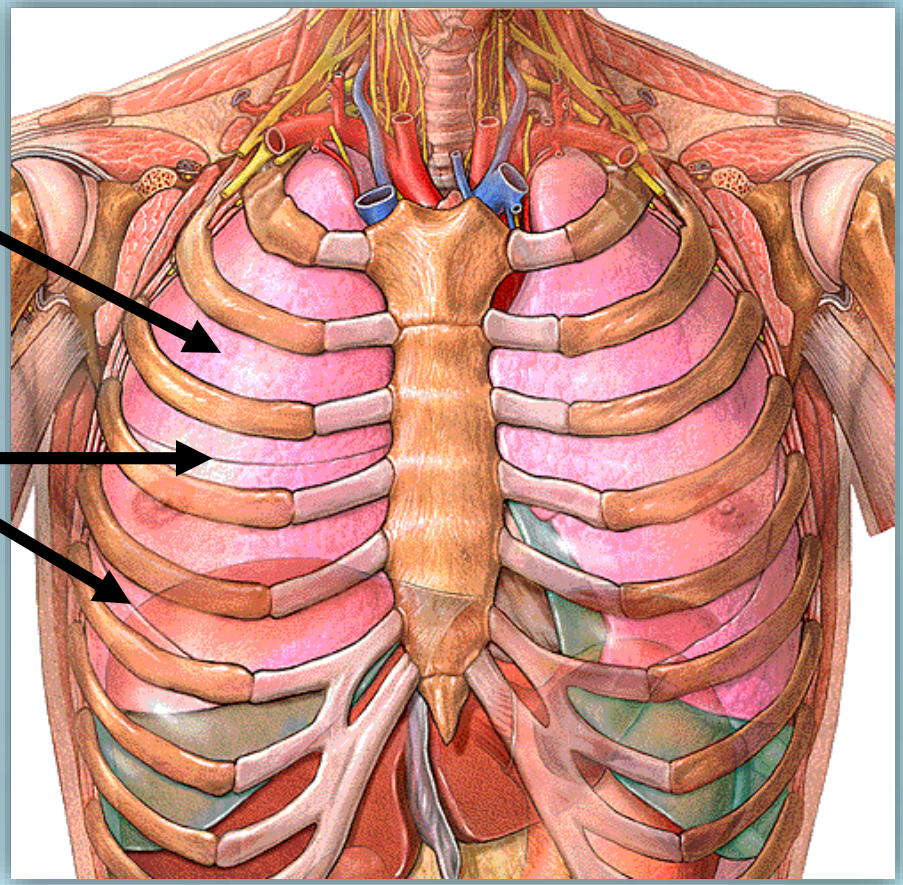
Poumon présente à décrire:

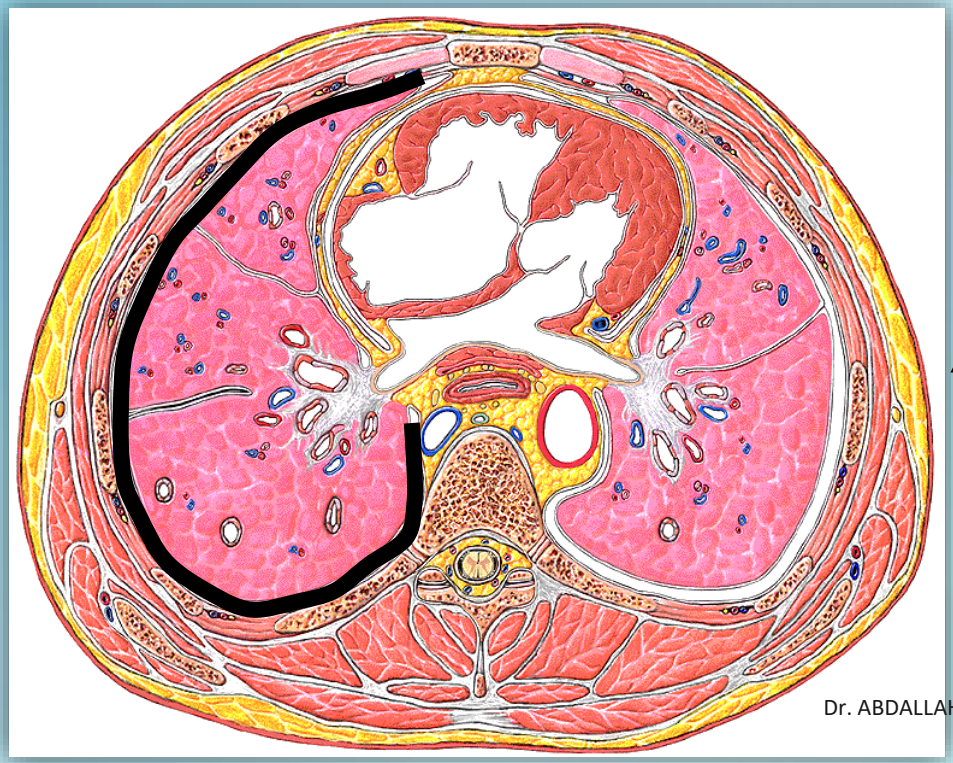
2 faces, **3 bords**, **1 base** et **1 apex**

Face latérale (costale):

Convexe

Traversée par: **scissures**





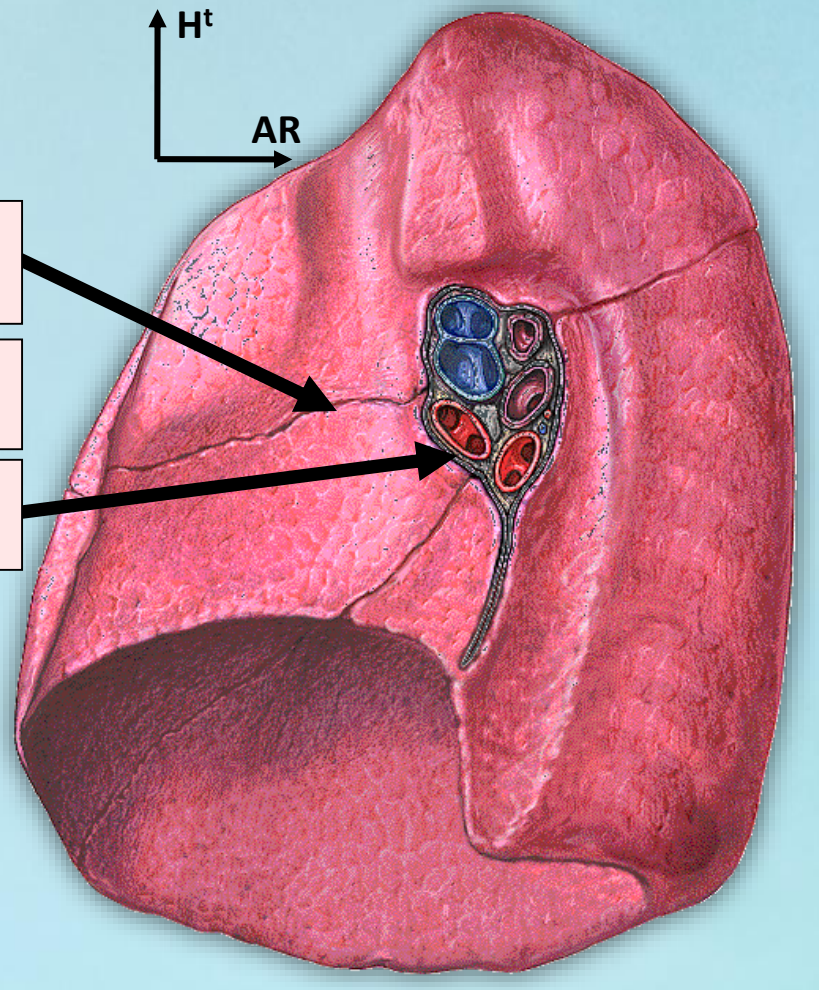
AV

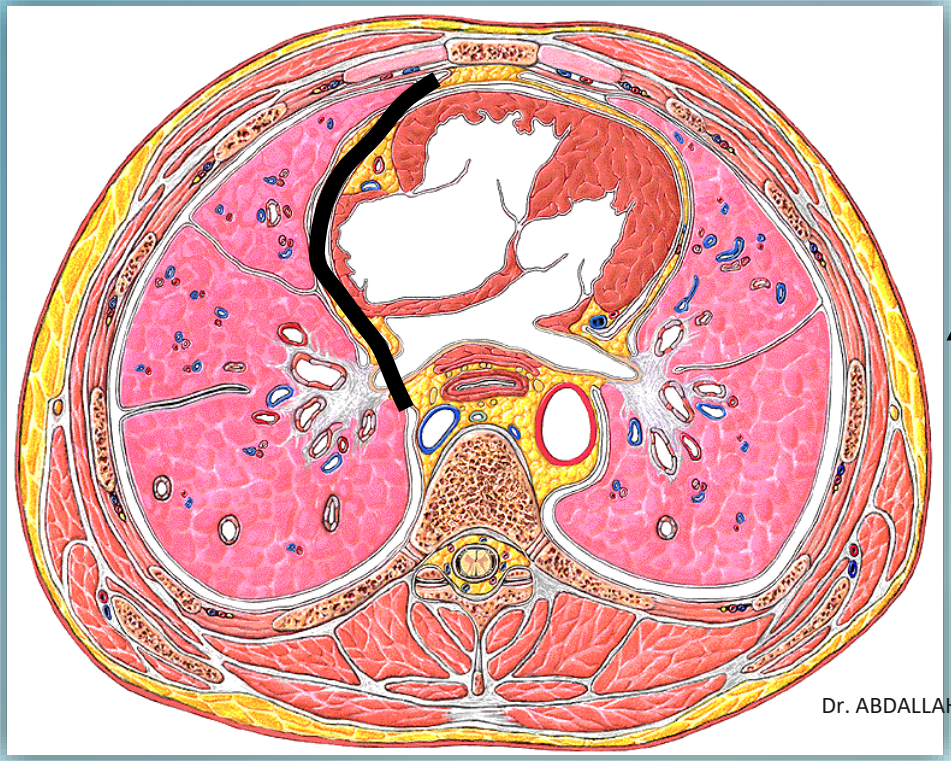
G^che

Face médiale (médiastinale):

Concave

Centrée par: **hile pulmonaire**



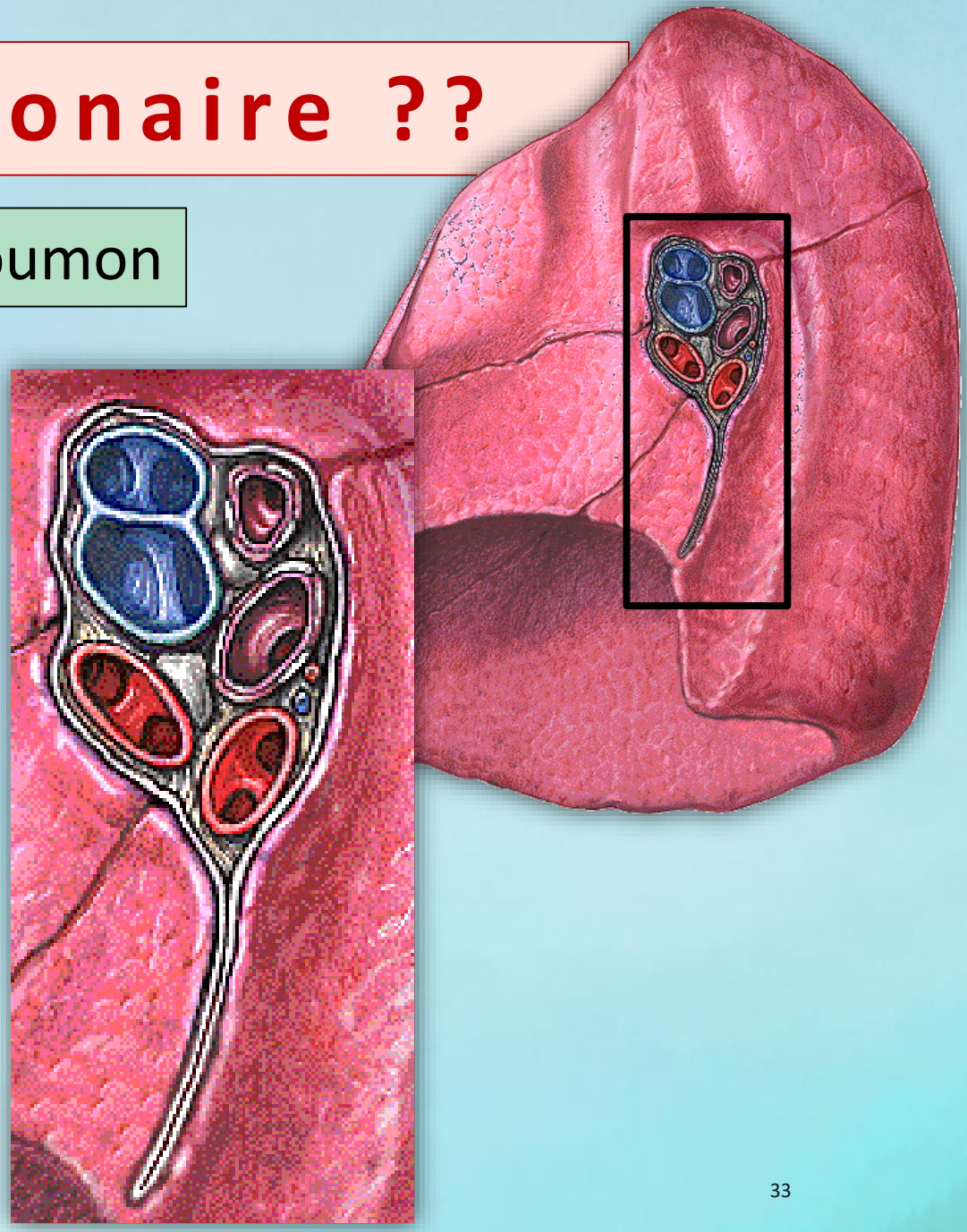


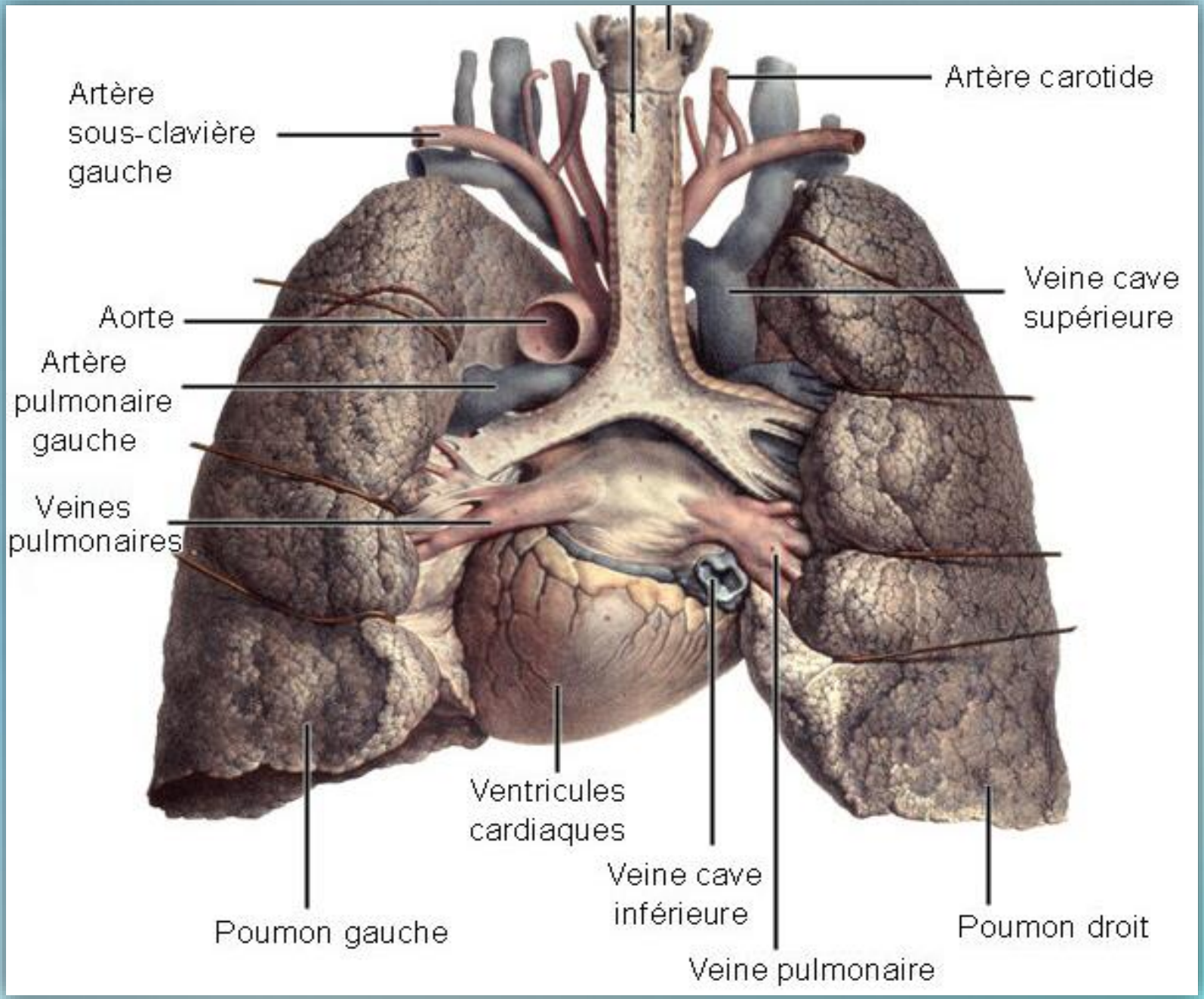
Hile pulmonaire ??

= **porte d'entrée** au poumon

Forme:
raquette

Éléments de passage:
pédicule pulmonaire +++

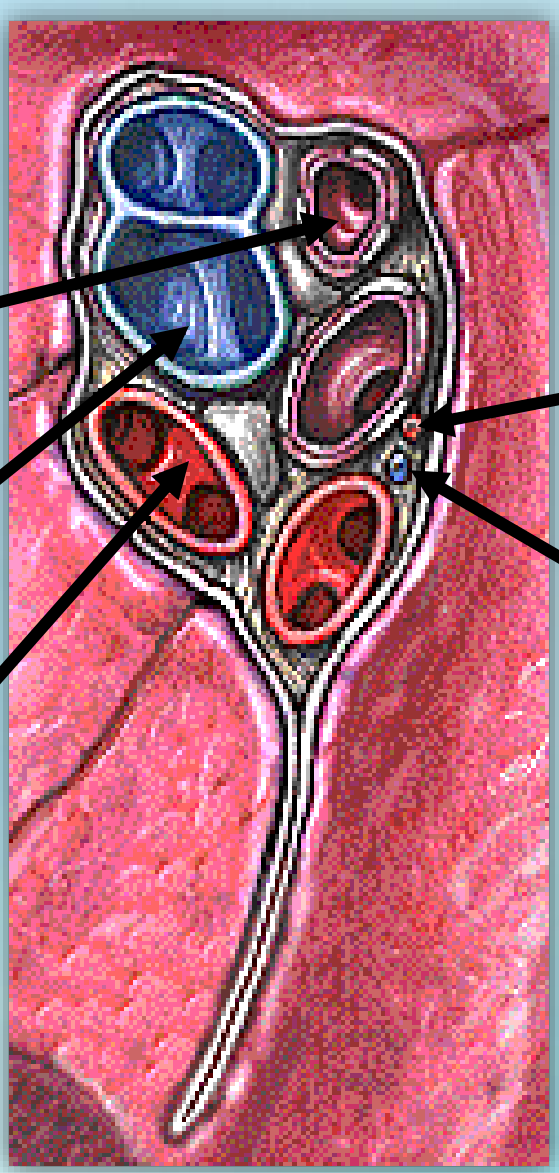



Artère carotide
Artère sous-clavière gauche
Veine cave supérieure
Aorte
Artère pulmonaire gauche
Veines pulmonaires
Ventricules cardiaques
Veine cave inférieure
Poumon gauche
Veine pulmonaire
Poumon droit

Pédicule pulmonaire ??

Éléments fonctionnels

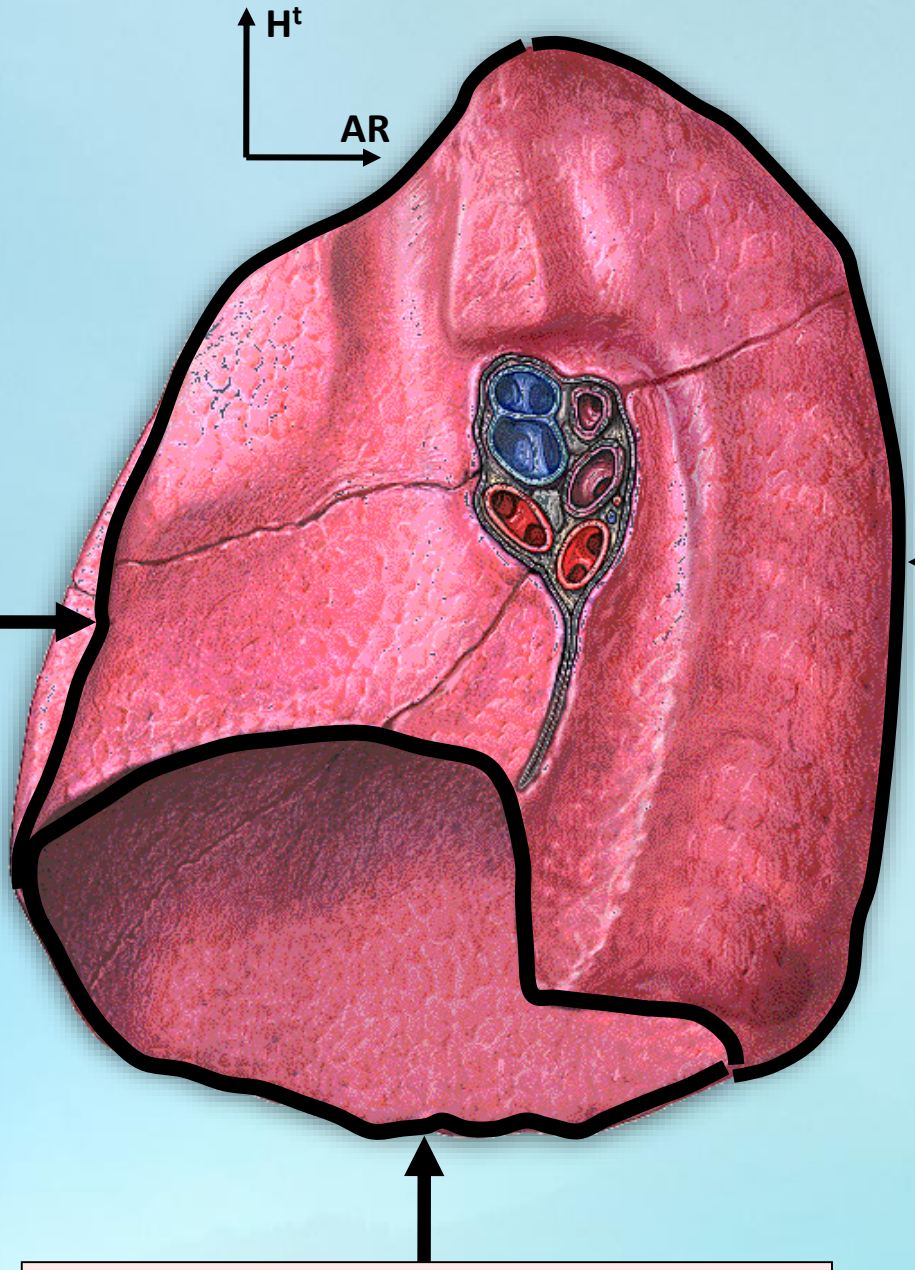
- **Bronches principales**
- **Artères pulmonaires** droite et gauche
- **Veines pulmonaires** droites et gauches



Éléments nourriciers:

- **Artères bronchiques**
- **Veines bronchiques**
- **Lymphatiques** bronchiques
- **Nerfs** bronchiques

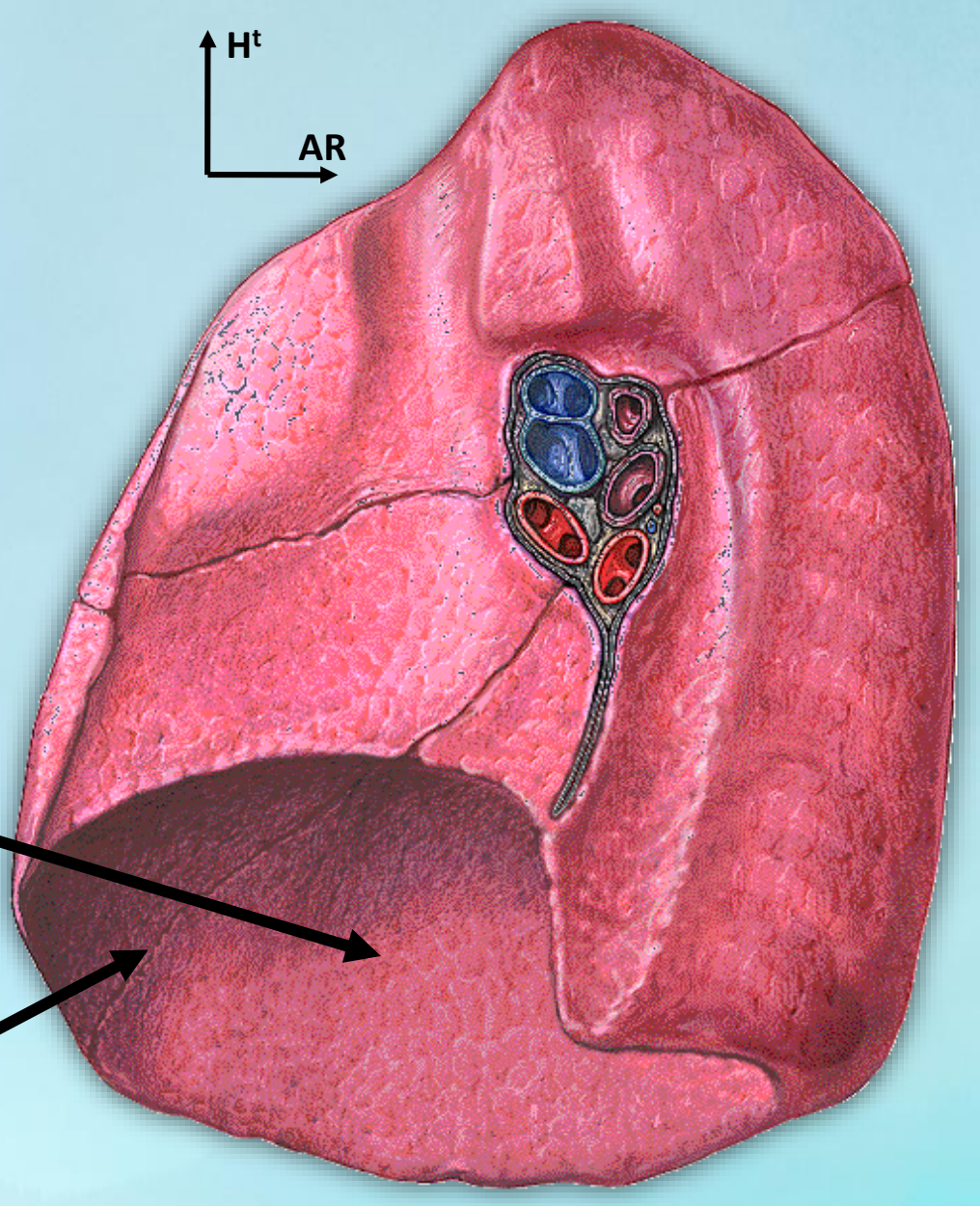
Bords:



Antérieur

Postérieur

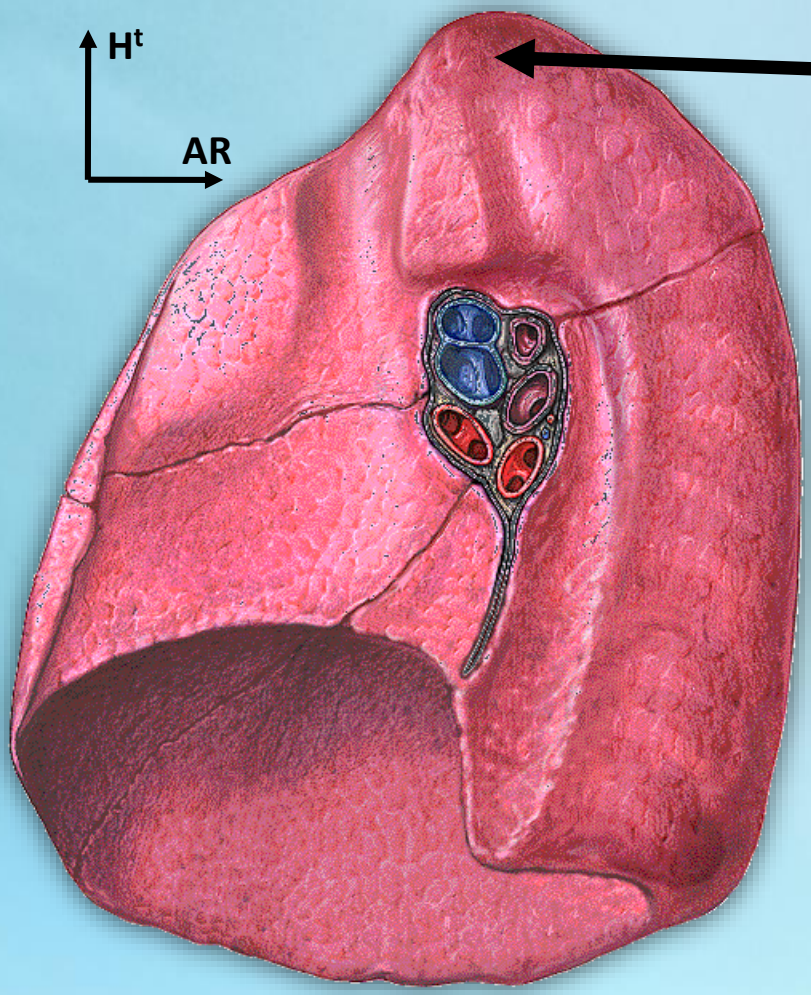
Inférieur (circonférentiel)



Base:

Inférieure et concave

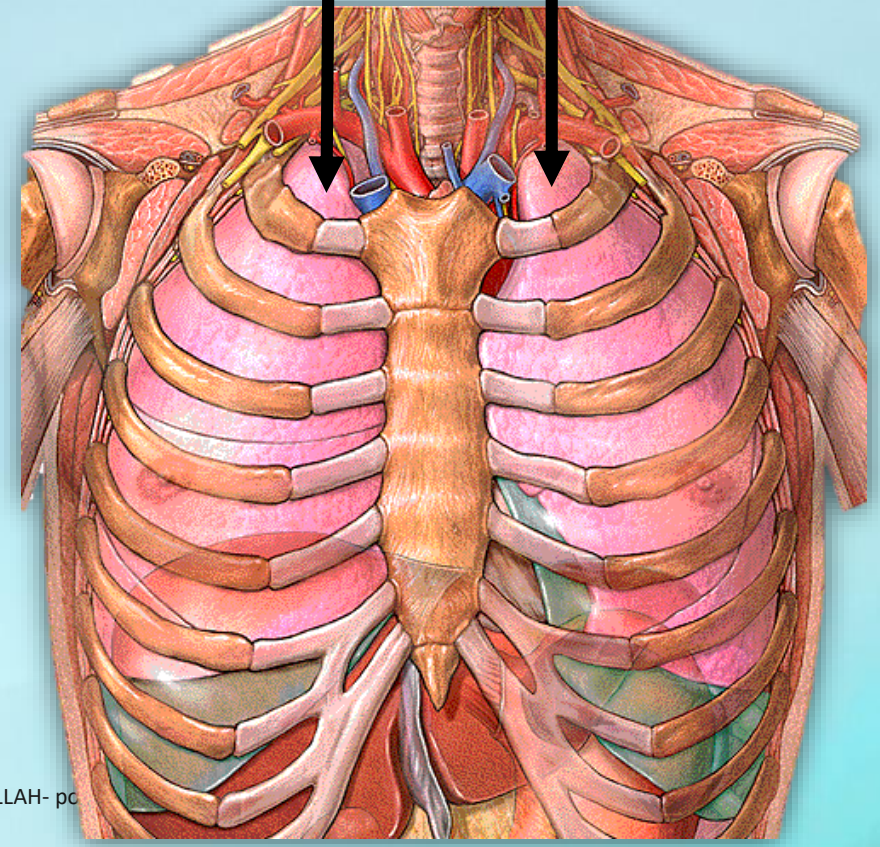
Traversée par:
scissure oblique



Apex:

supérieur

Déborde en haut
l'orifice supérieur du thorax



Division & Segmentation

3 scissures

divisent les **poumons**

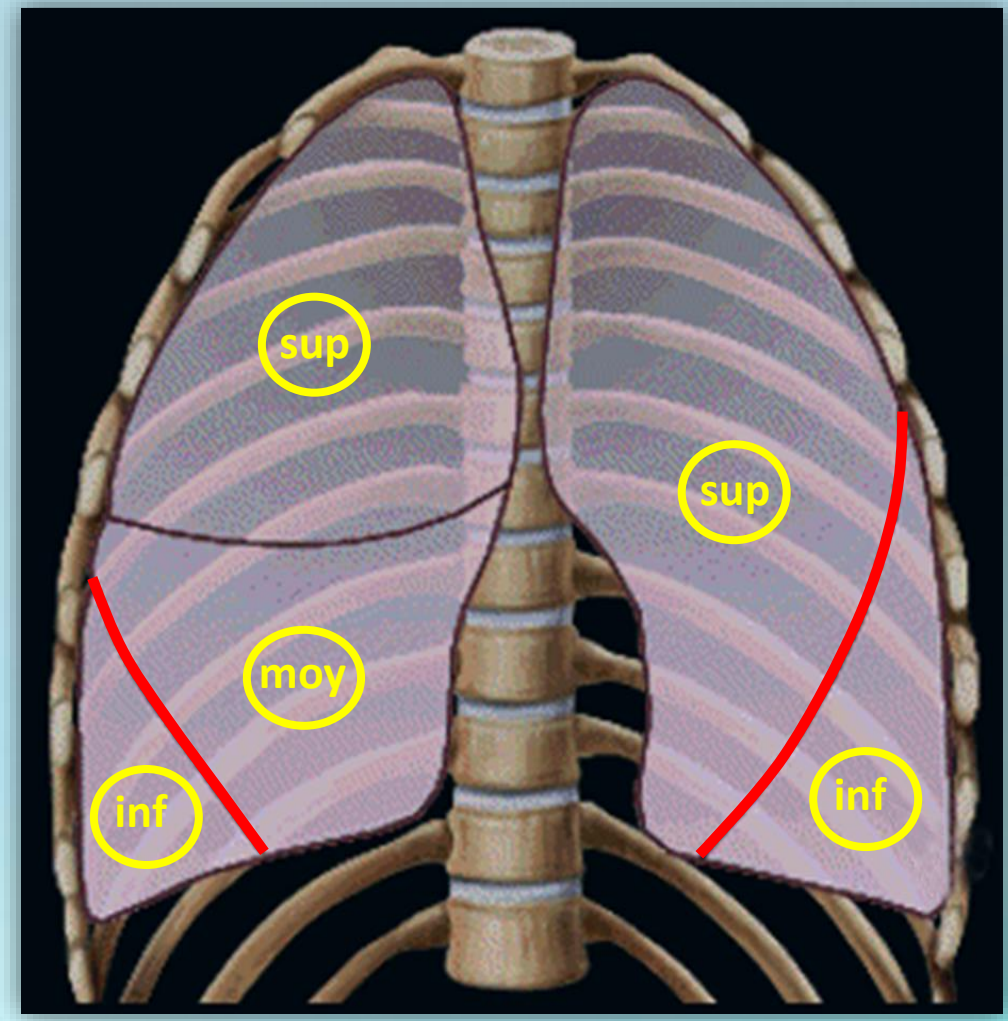
en **5 lobes..**

Scissure oblique

- **visible** sur les **2 poumons**
- **oblique** en avant et en bas

- divise <u>poumon gauche</u> en **2 lobes:** **supérieur** et **inférieur**

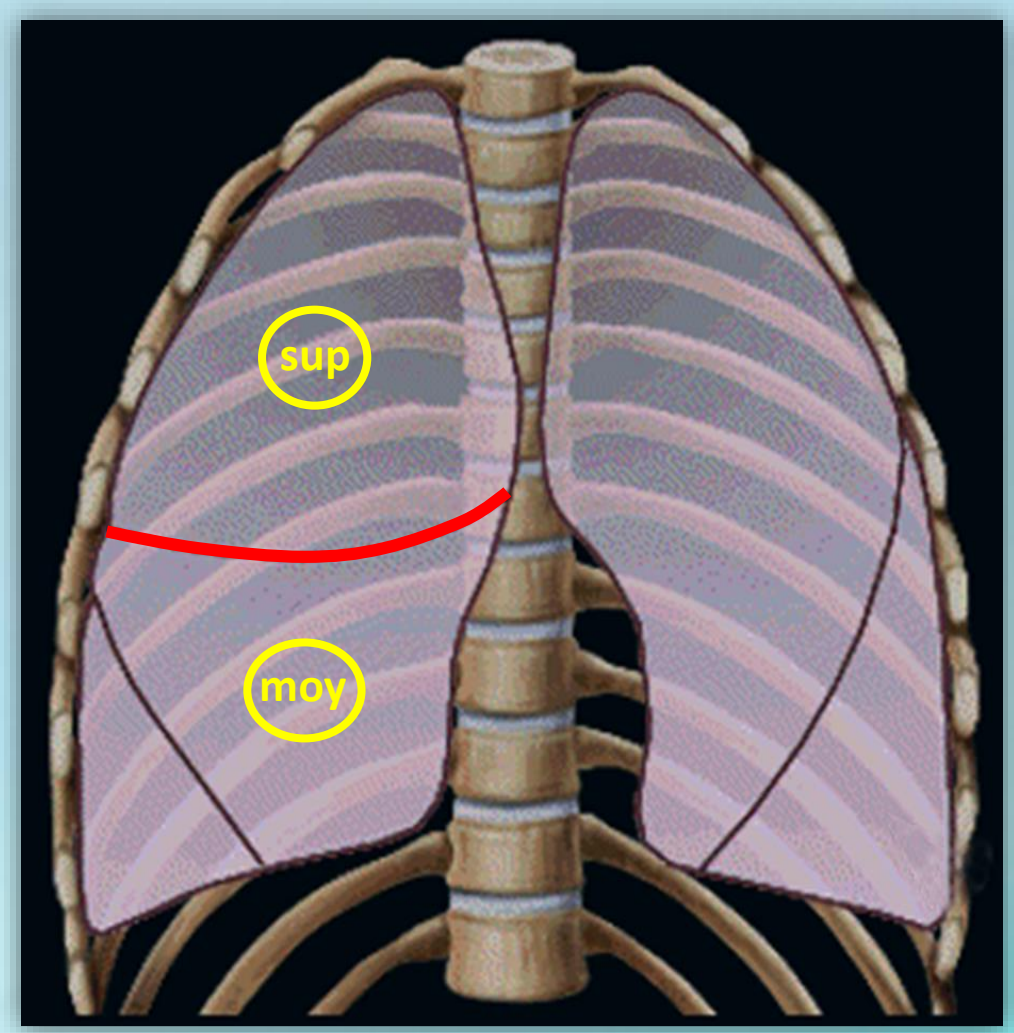
- sur <u>poumon droit</u>, sépare **lobe inférieur** des **2 lobes supérieur** et **moyen**

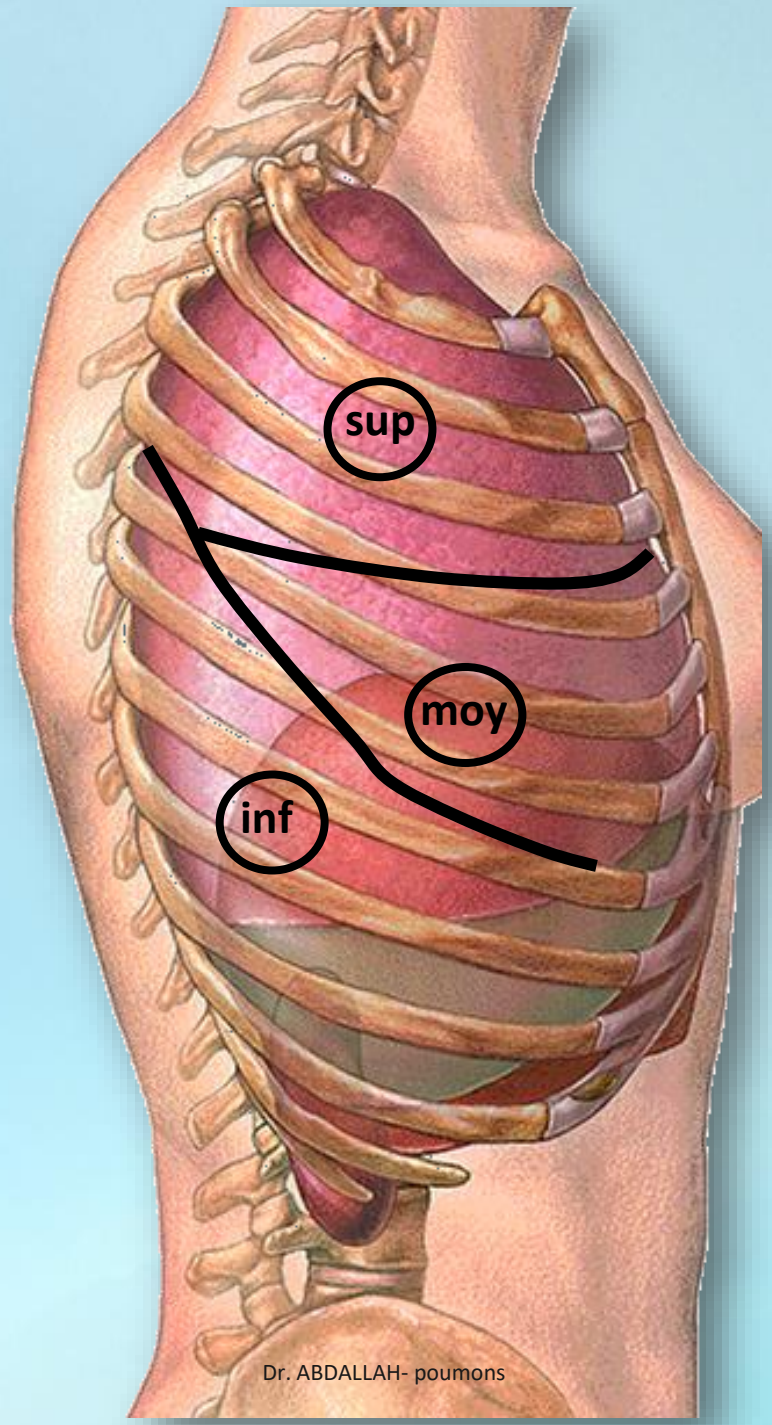


Scissure horizontale

- **propre** au **poumon droit**
- naît de **la scissure oblique**

- sépare: **lobe supérieur** du **lobe moyen**

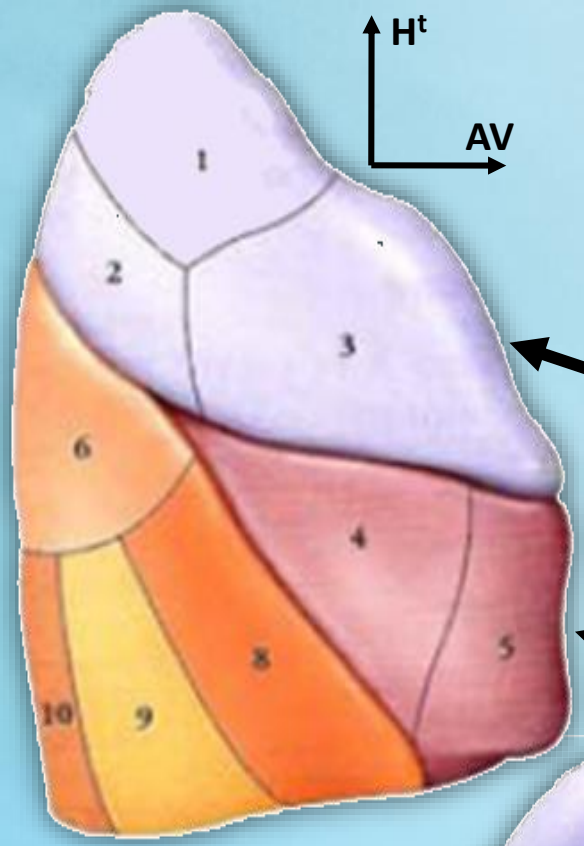


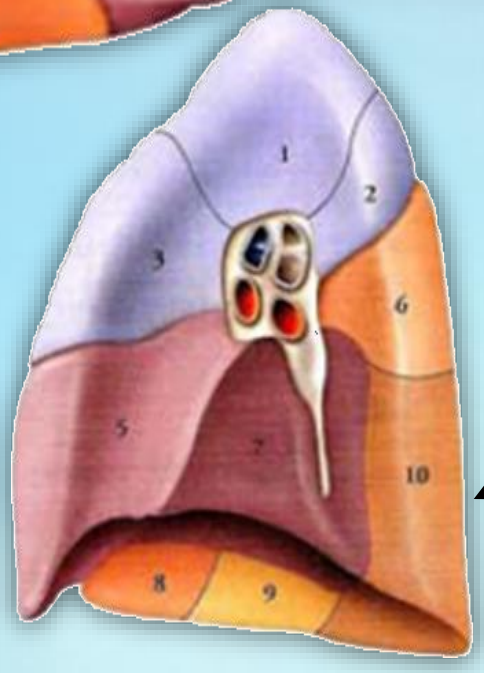

sup
moy
inf

Chaque **lobe pulmonaire**

se subdivise en **segments**

Poumon droit





Lobe supérieur: 3 segments
apical (1), postérieur (2), antérieur (3)

Lobe moyen: 2 segments
latéral (4), médial (5)

Lobe inférieur: 5 segments
apical (6), basal médial (7), basal antérieur (8), basal latéral (9), basal postérieur (10)

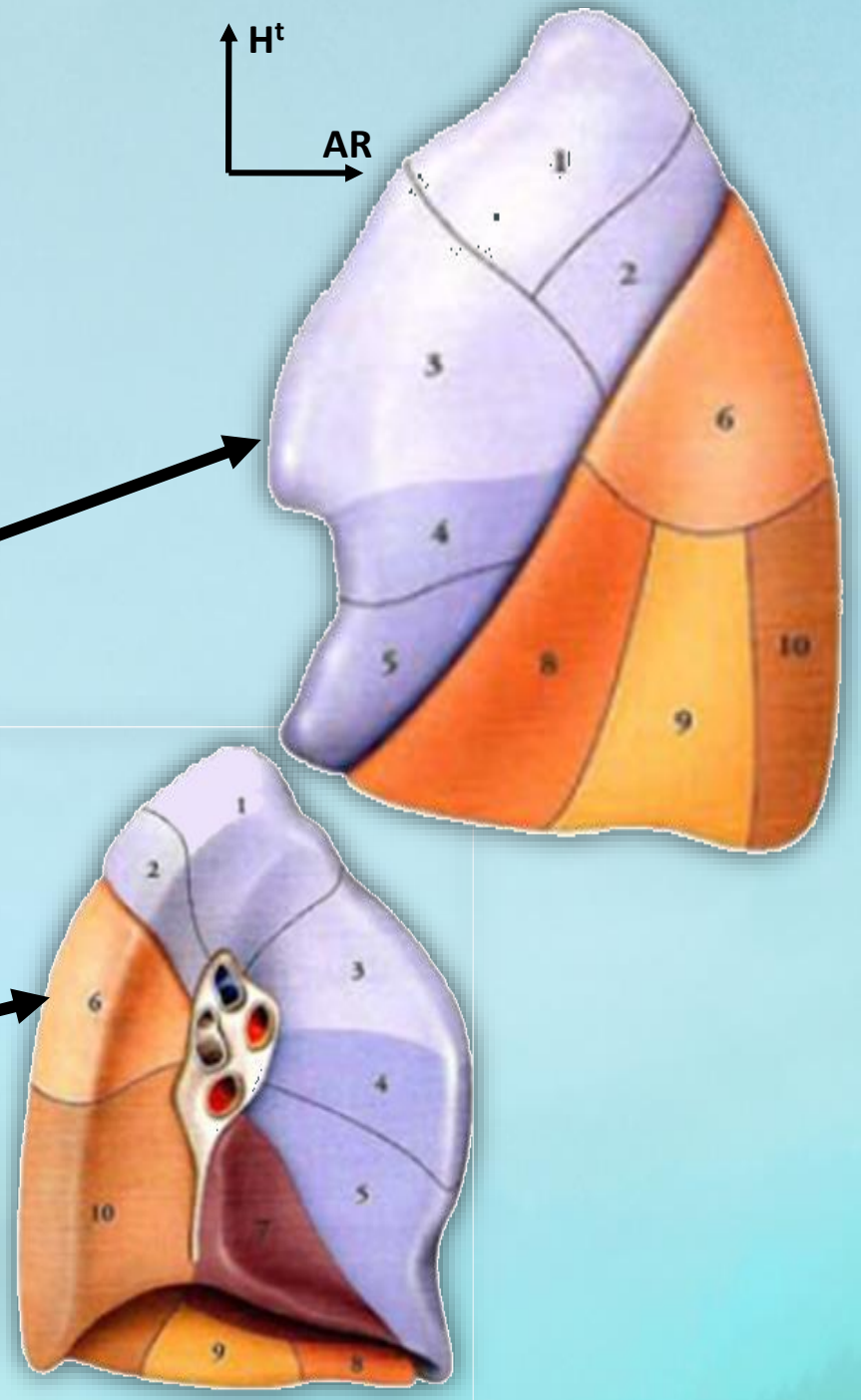
Poumon gauche

Lobe supérieur: 4 segments

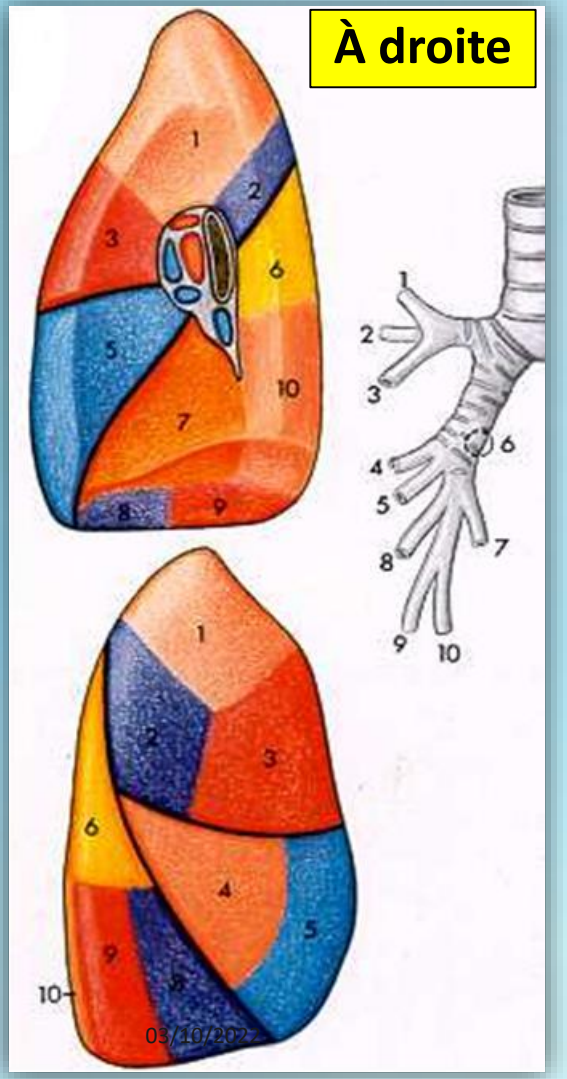
Apico-postérieur (1, 2), antérieur (3), lingulaire supérieur (4), lingulaire inférieur (5)

Lobe inférieur: 5 segments

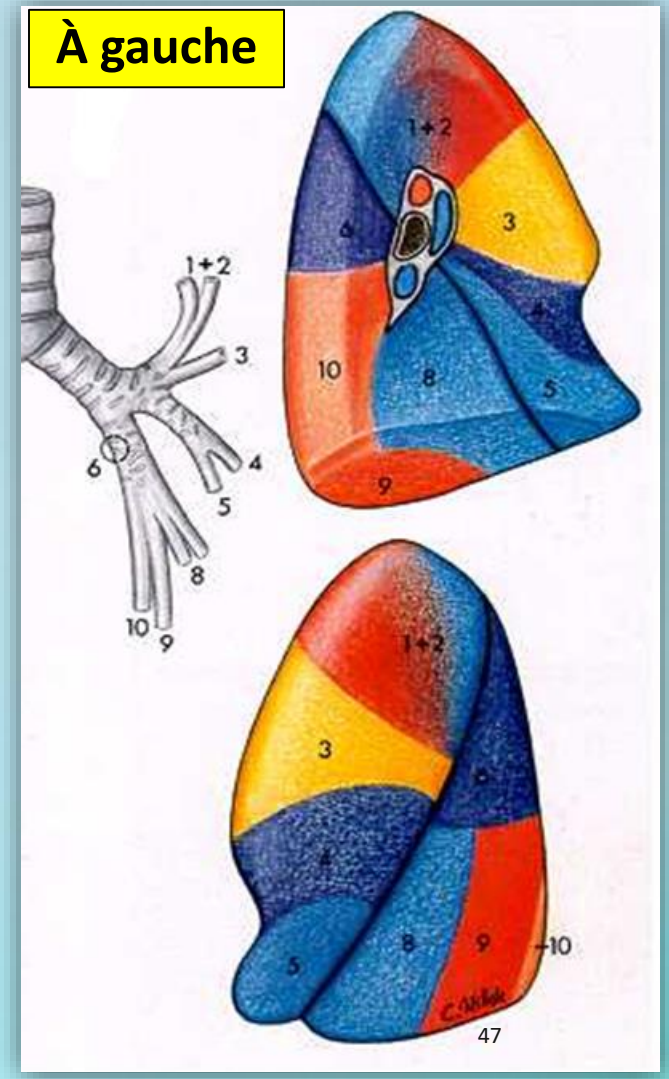
apical (6), basal médial (7), basal antérieur (8), basal latéral (9), basal postérieur (10)



Remarque:



Les 2 segmentations, pulmonaire et bronchique se suivent



Poumon

Lobes

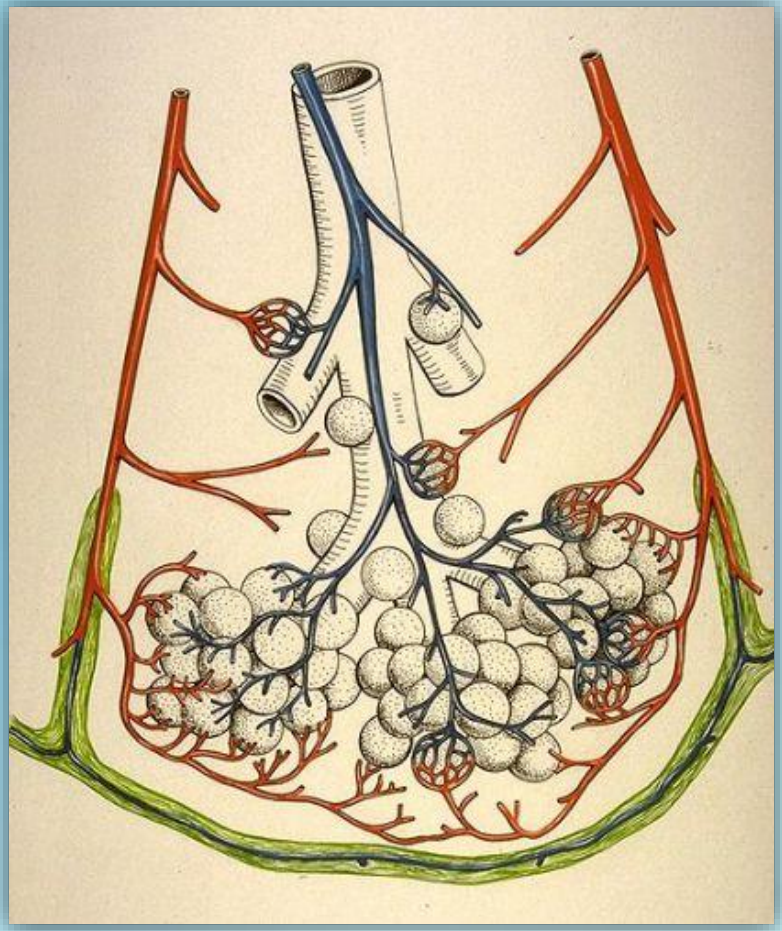
Segments

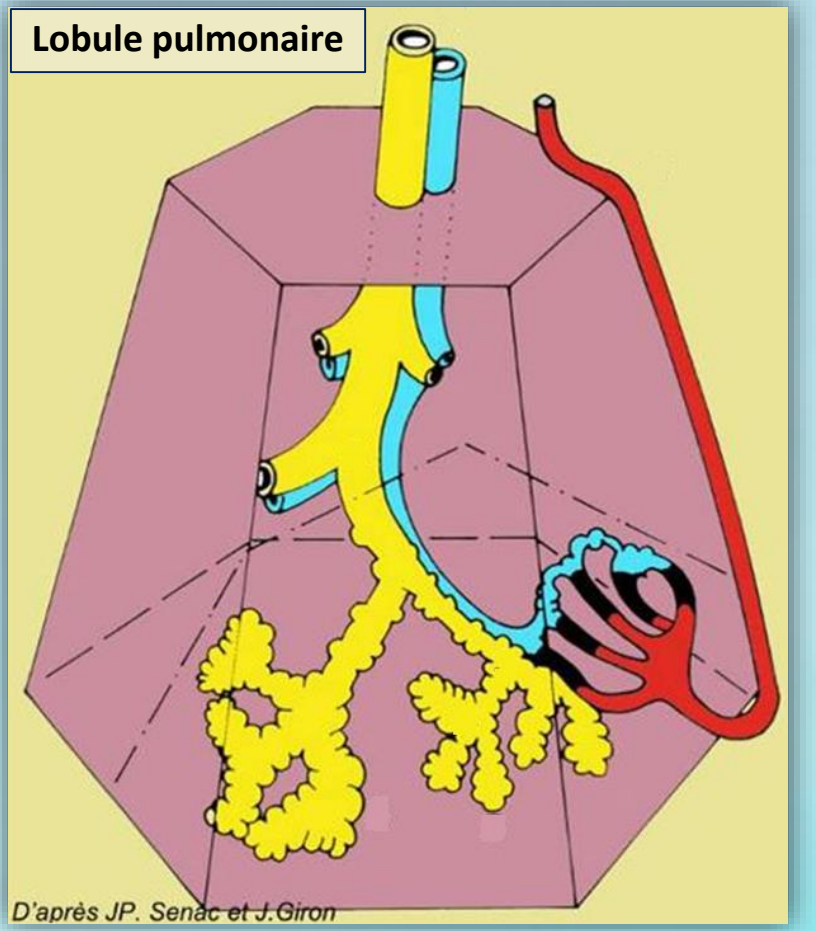
Chaque **segment pulmonaire**
se subdivise en: **sous-segments**

Chaque **sous-segment pulmonaire**
se subdivise en: **territoires plus petits**
.. et ainsi de suite…

Arrivant, **à la fin**, à de petites masses
(volume = 1 cc): **lobules pulmonaires**

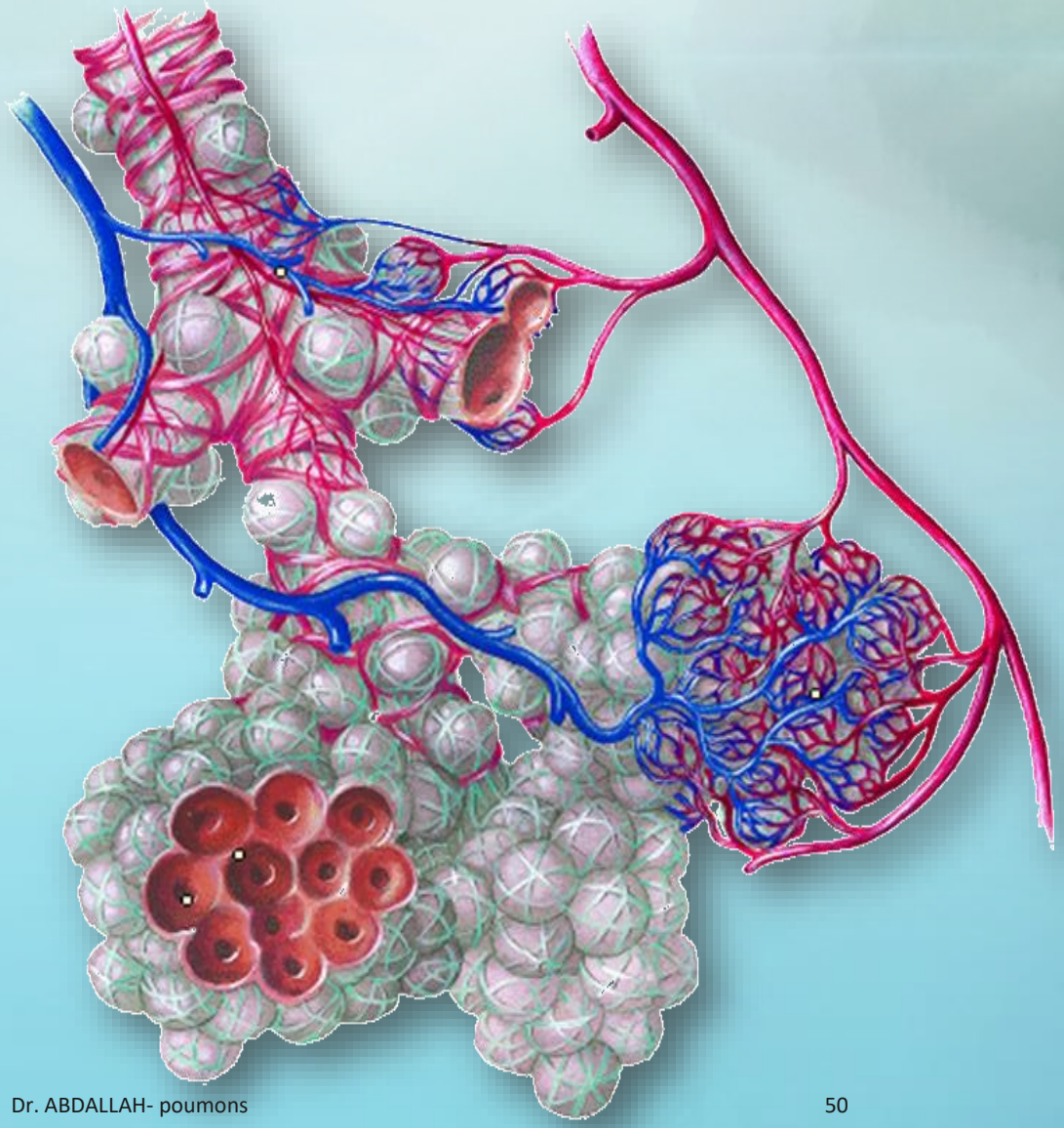
Chaque **lobule pulmonaire** contient un ensemble **d'alvéoles**



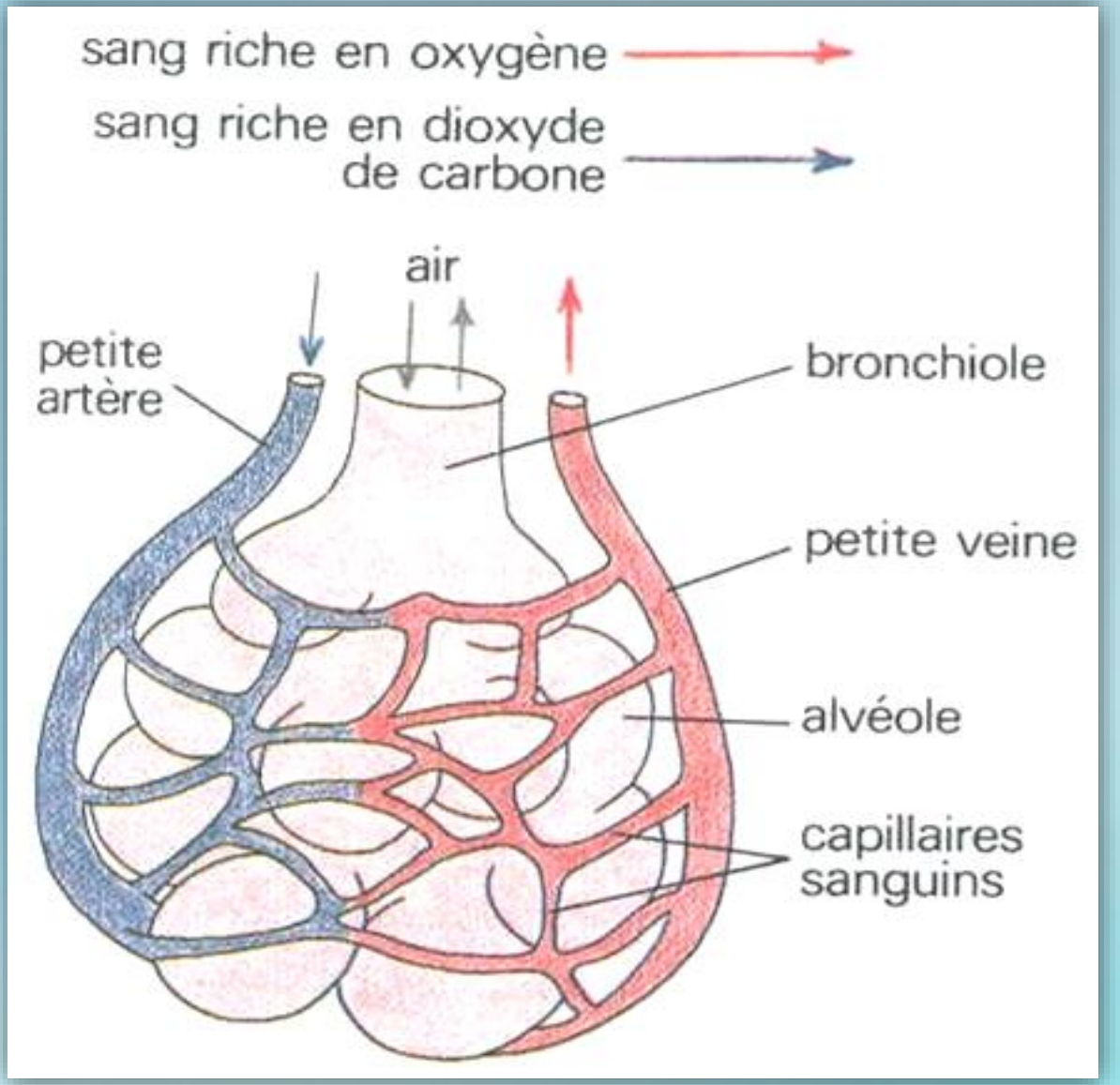


Alvéoles pulmonaires ??

- Unités fonctionnelles des poumons
- = **minuscules sacs**, tapissés par des **capillaires**
- Nombre = **300 millions**
- Où s'effectuent: **échanges gazeux**

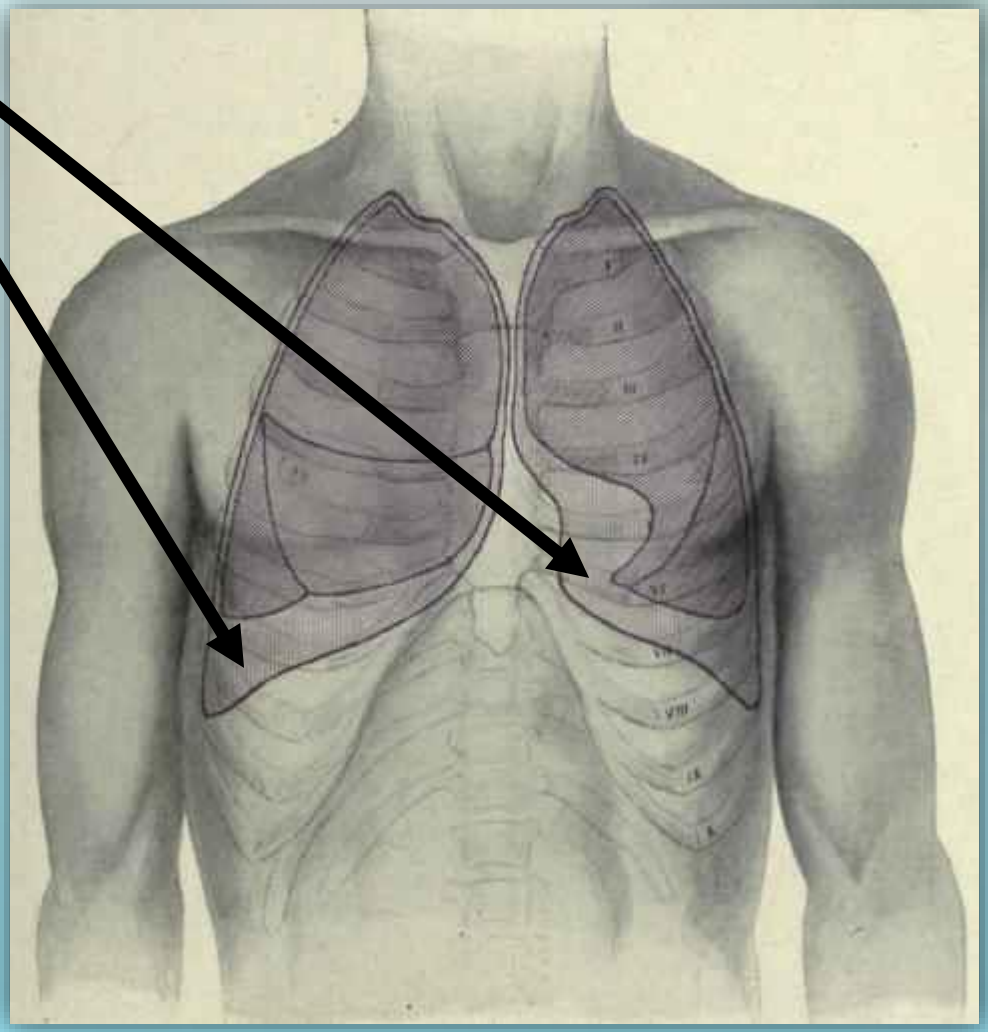


Comment s'effectuent les échanges gazeux ?



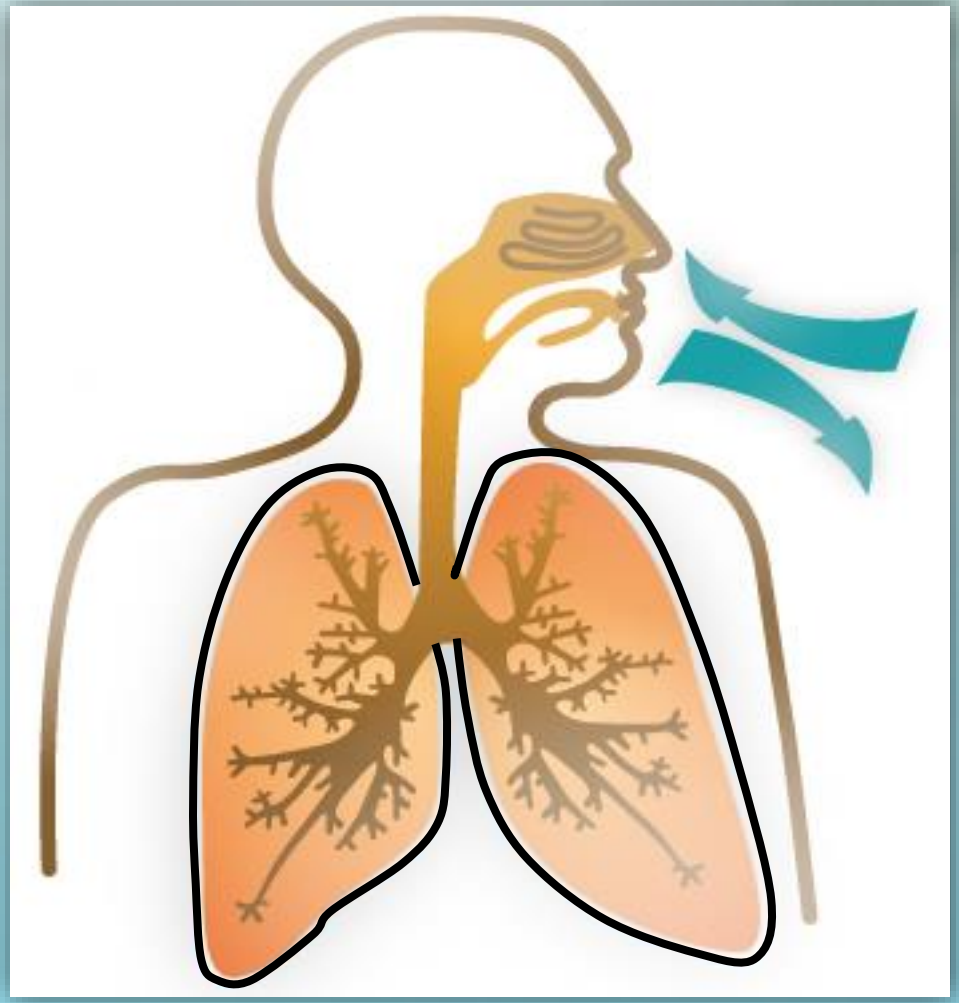

Plèvres

- = **2 membranes séreuses**
- **Indépendantes** l'une de l'autre
- Enveloppant: **poumons**
- Tapissant: face profonde des **cavités pleuro-pulmonaires**



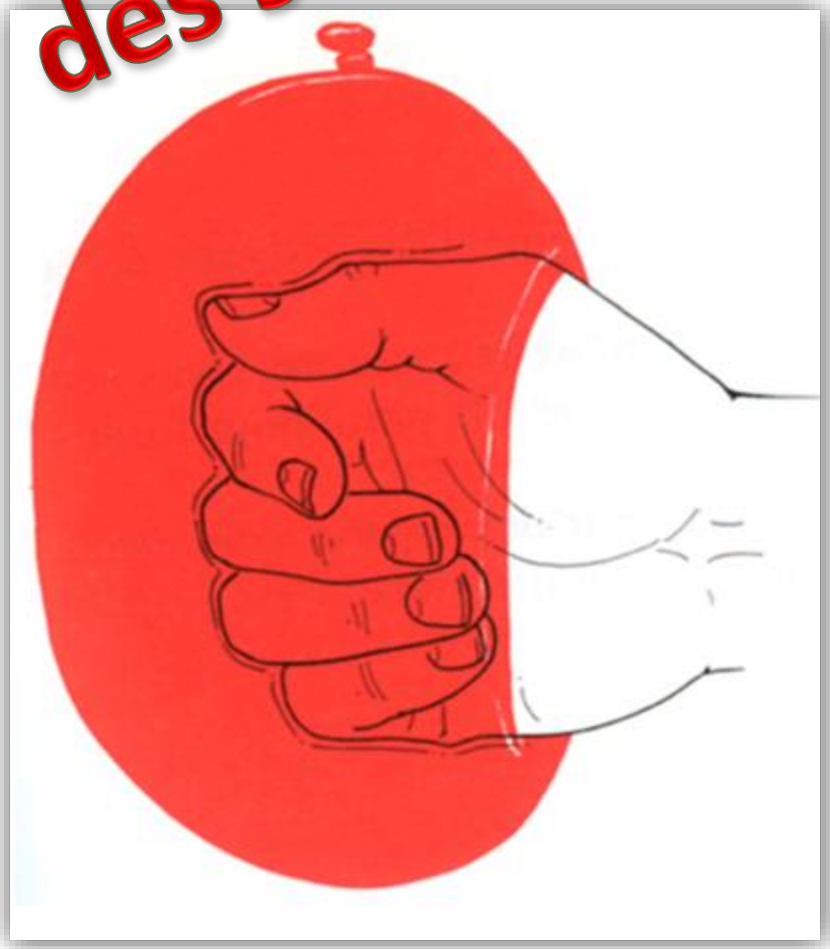
- **Rôle:**

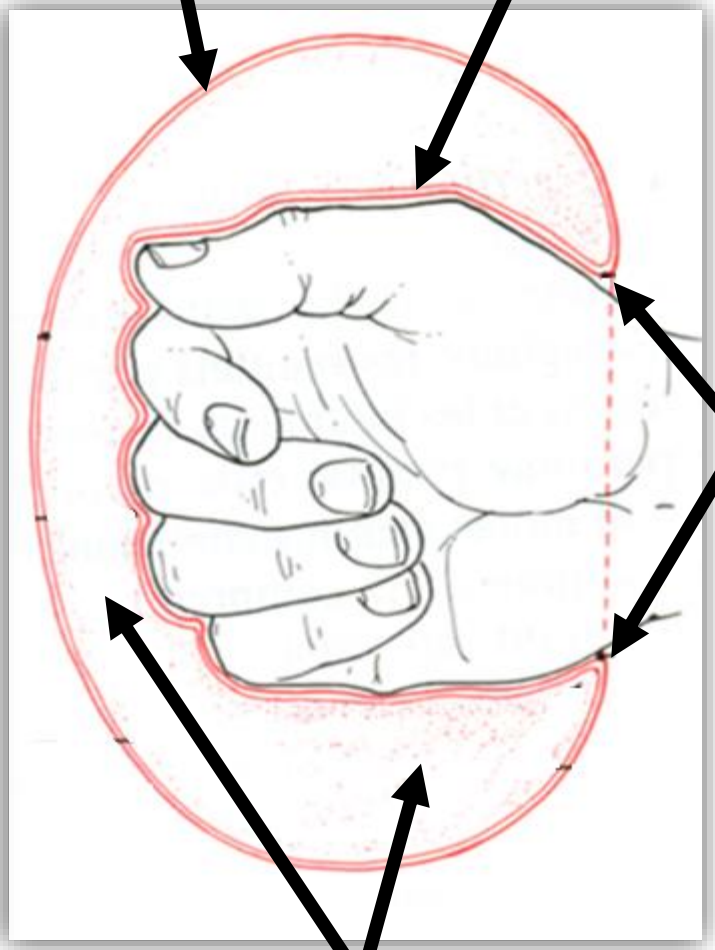
  Faciliter **glissement** des **poumons** sur **paroi thoracique**

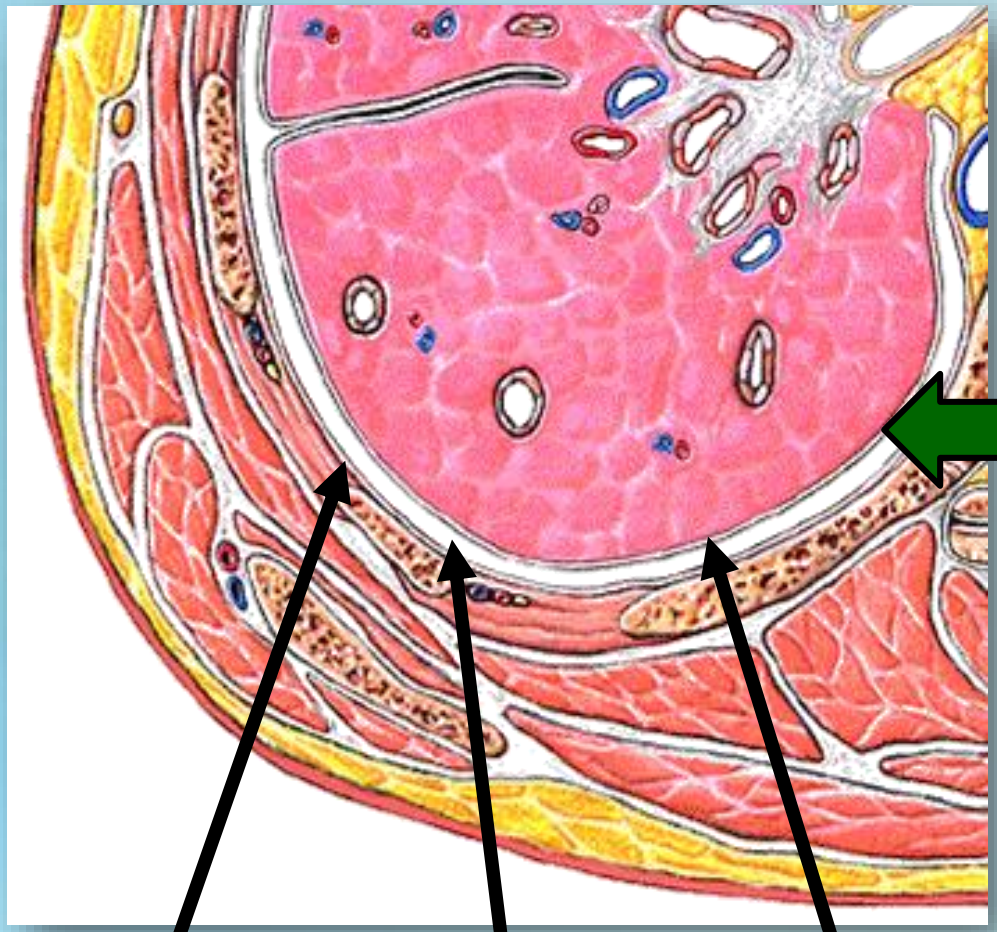


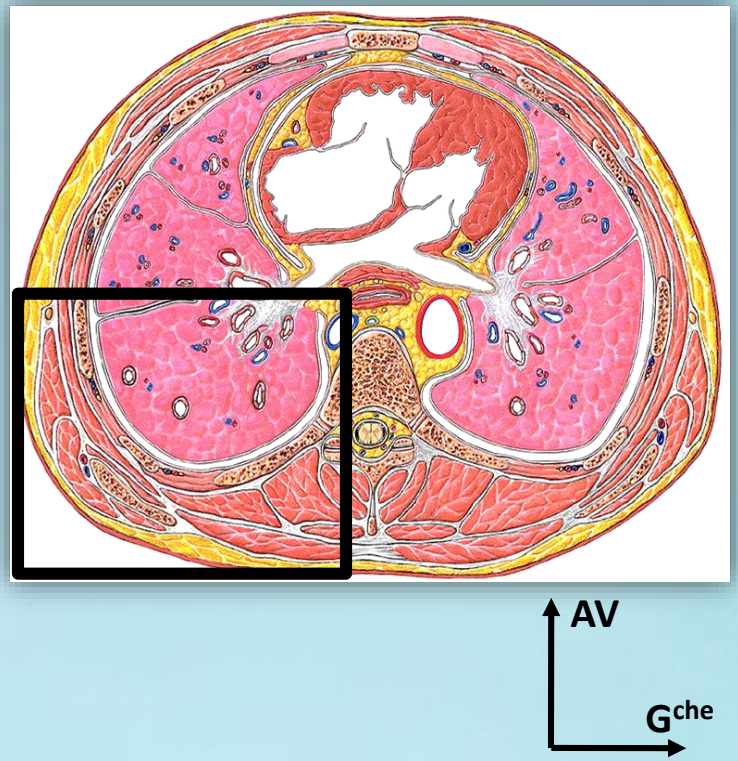
Constitution

Concept des séreuses









Feuillet pariétal

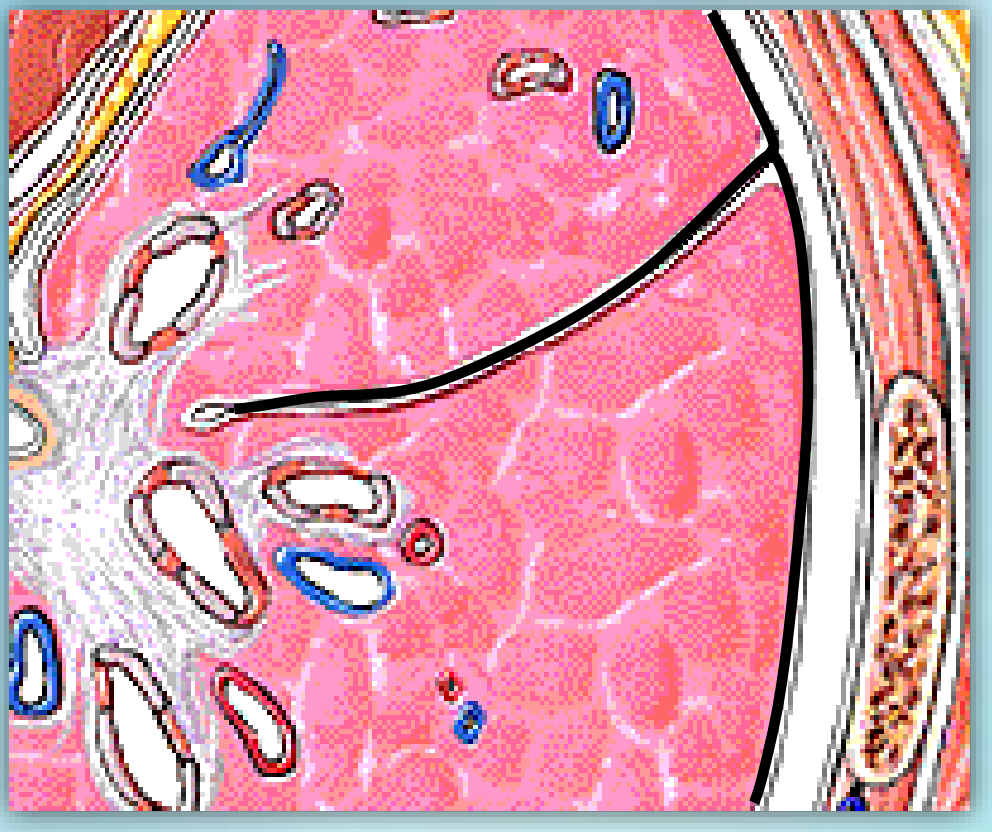
Cavité pleurale

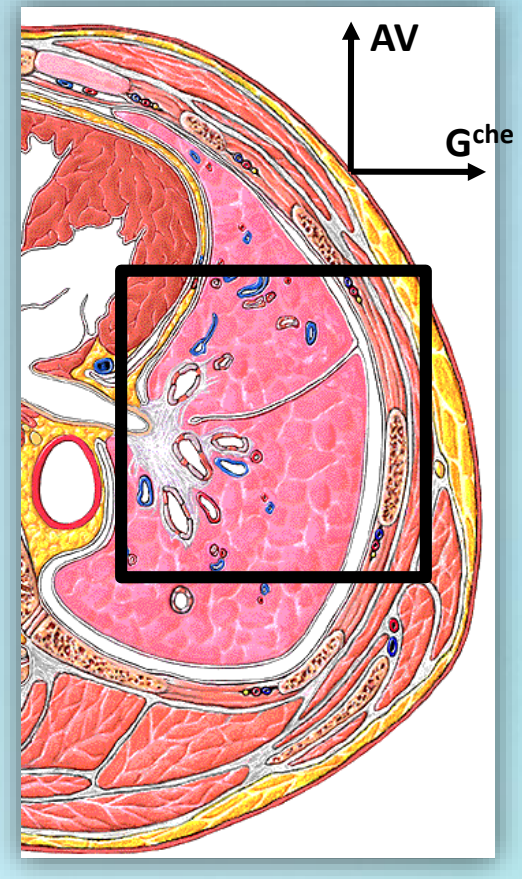
Feuillet viscéral

Feuillet viscéral

Tapisse: **surface du poumon**

S’enfonce dans:
scissures pulmonaires

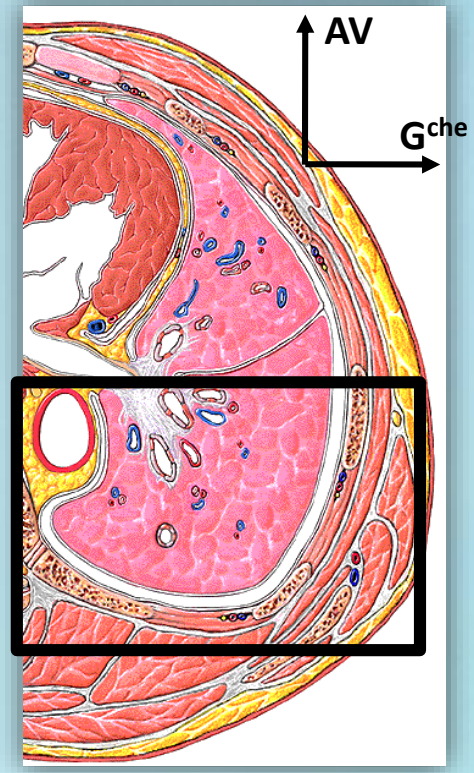


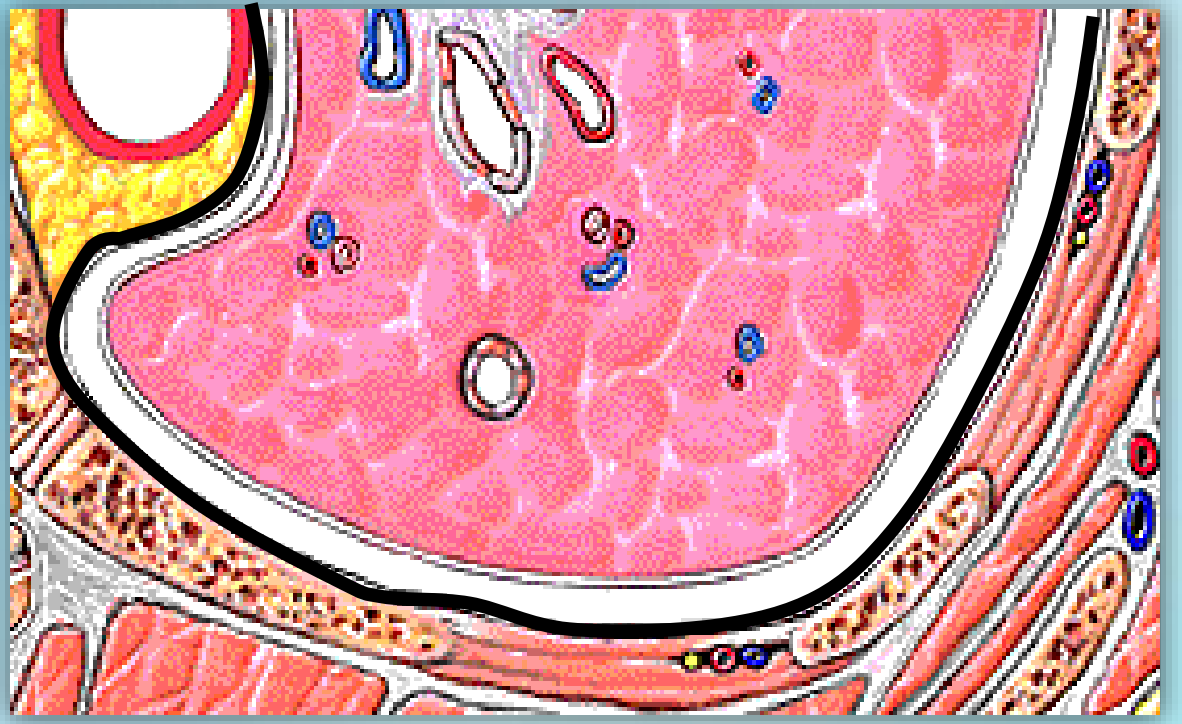


Feuillet pariétal

Tapisse:

face profonde des cavités pleuro-pulmonaires



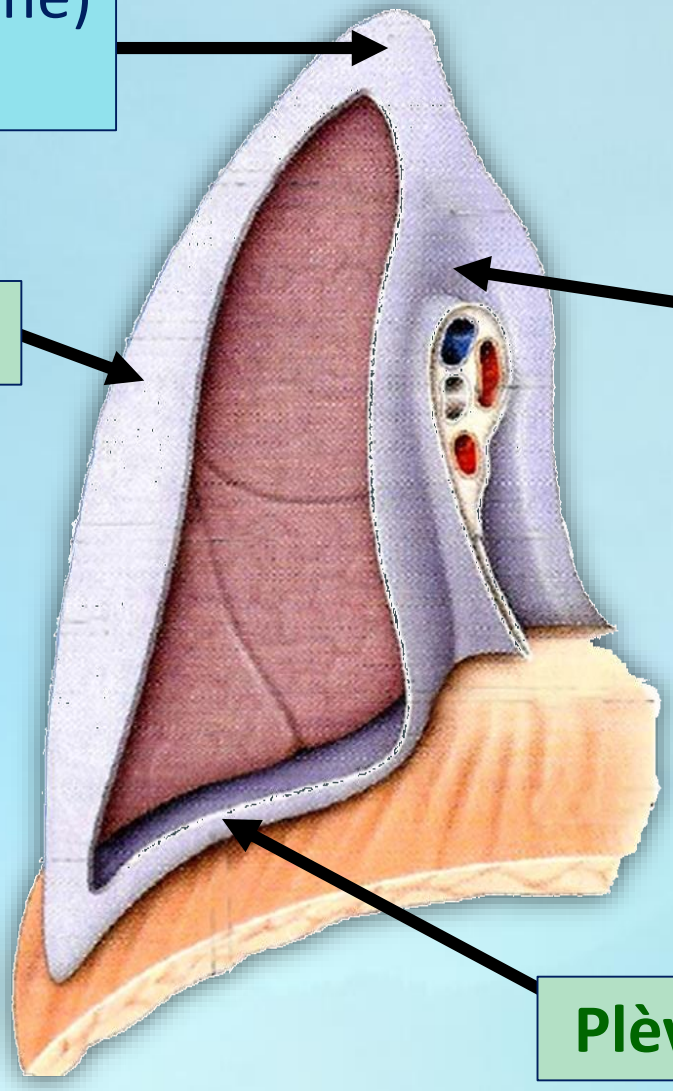


Présente:

4 segments topographiques

Coupole (dôme) **pleurale**

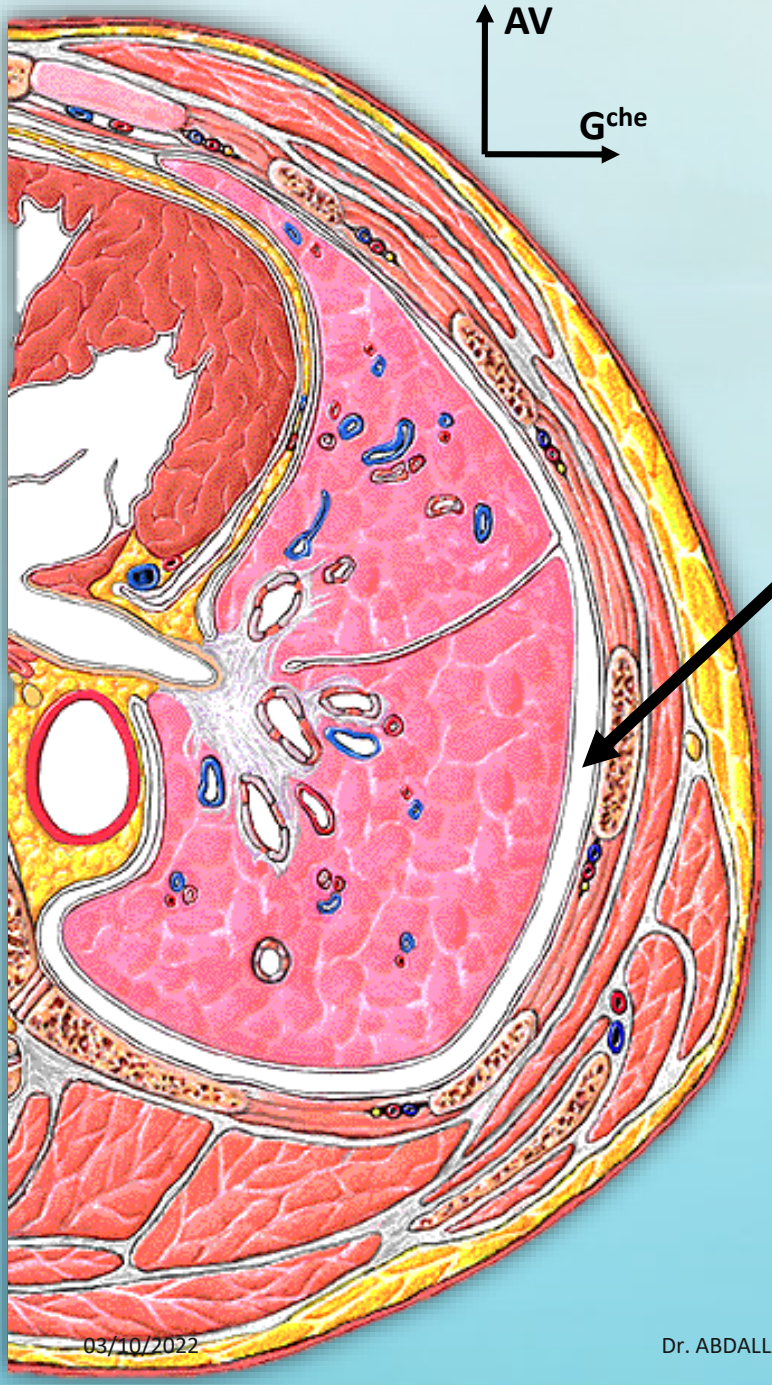
Plèvre costale



Plèvre médiastinale

Plèvre diaphragmatique

Cavité pleurale

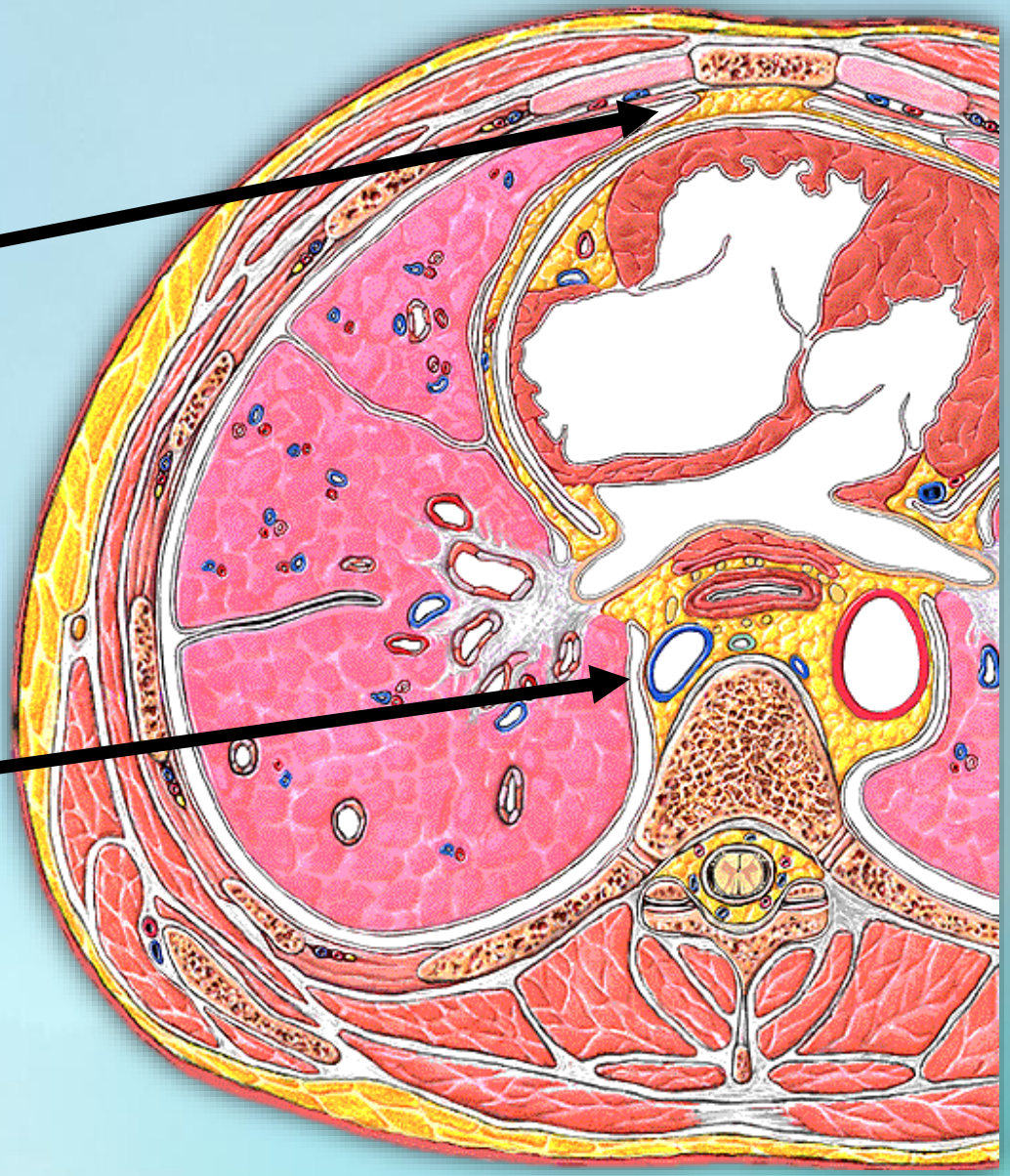


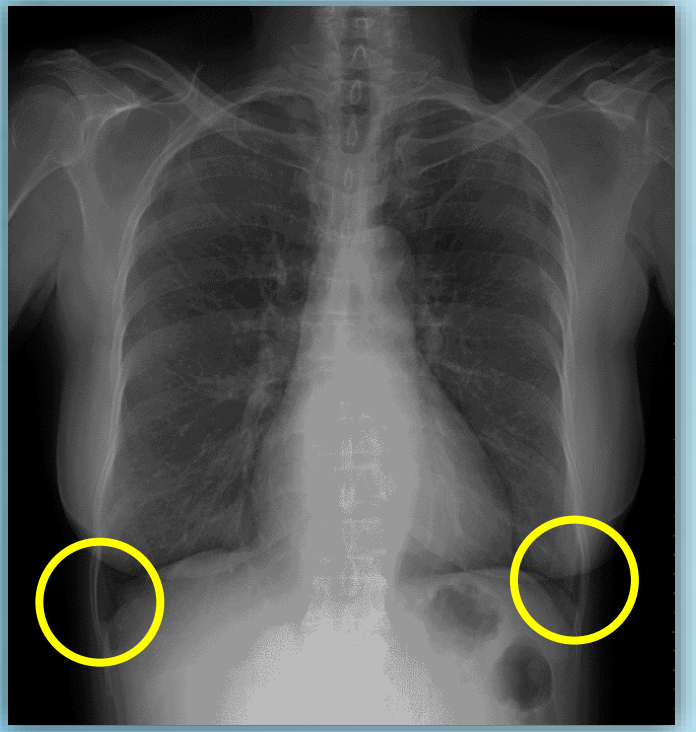
- **Cavité virtuelle**
- Située entre: **2 feuillets**
- Facilite:
  glissement des 2 feuillets
  lors des mouvements respiratoires

Présente: **4 récessus** (culs-de-sacs) **pleuraux**

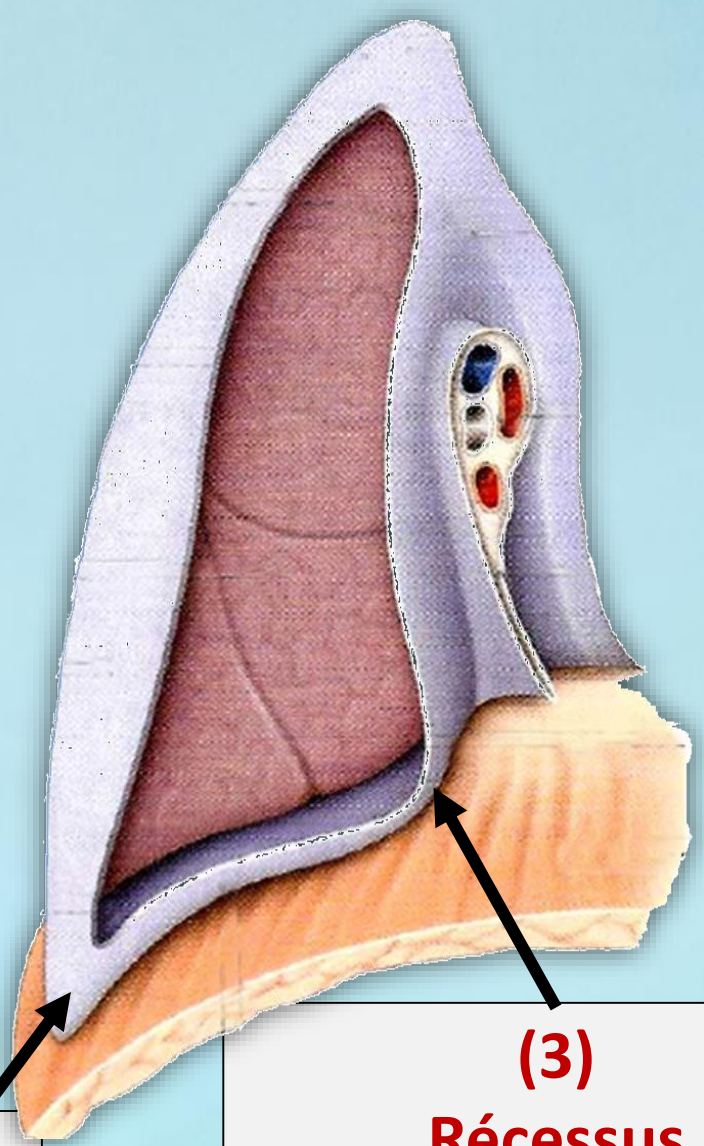
(1)
Récessus costo-médiastinal antérieur
(rétro-sternal)

(2)
Récessus costo-médiastinal postérieur
(latéro-vertébral)





Visible sur **téléthorax**



(3)
Récessus phrénico-médiastinal

(4)
Récessus costo-diaphragmatique
(inférieur)

Remarque:

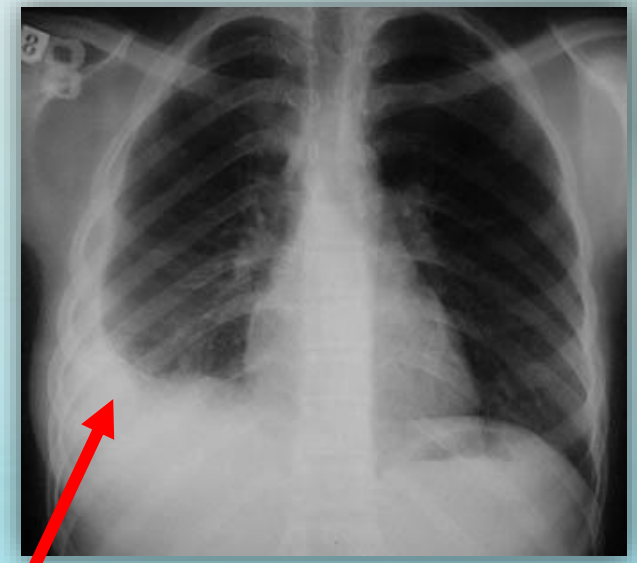
Épanchements de la cavité pleurale

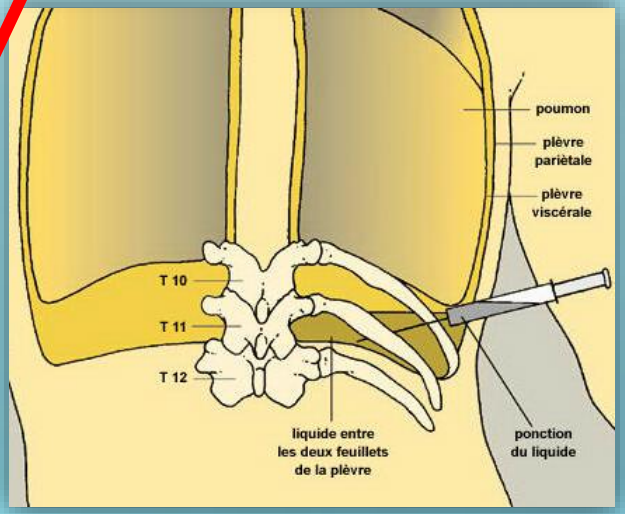
- Peuvent êtres:
  liquidiens (pleurésie, hémothorax)
  ou: **gazeux** (pneumothorax)

  ➔ Cavité pleurale **n'est plus virtuelle**

- Récessus costo-diaphragmatique peut **se combler ➔ s'effacer**

  ➔ On pratique: **ponction pleurale**

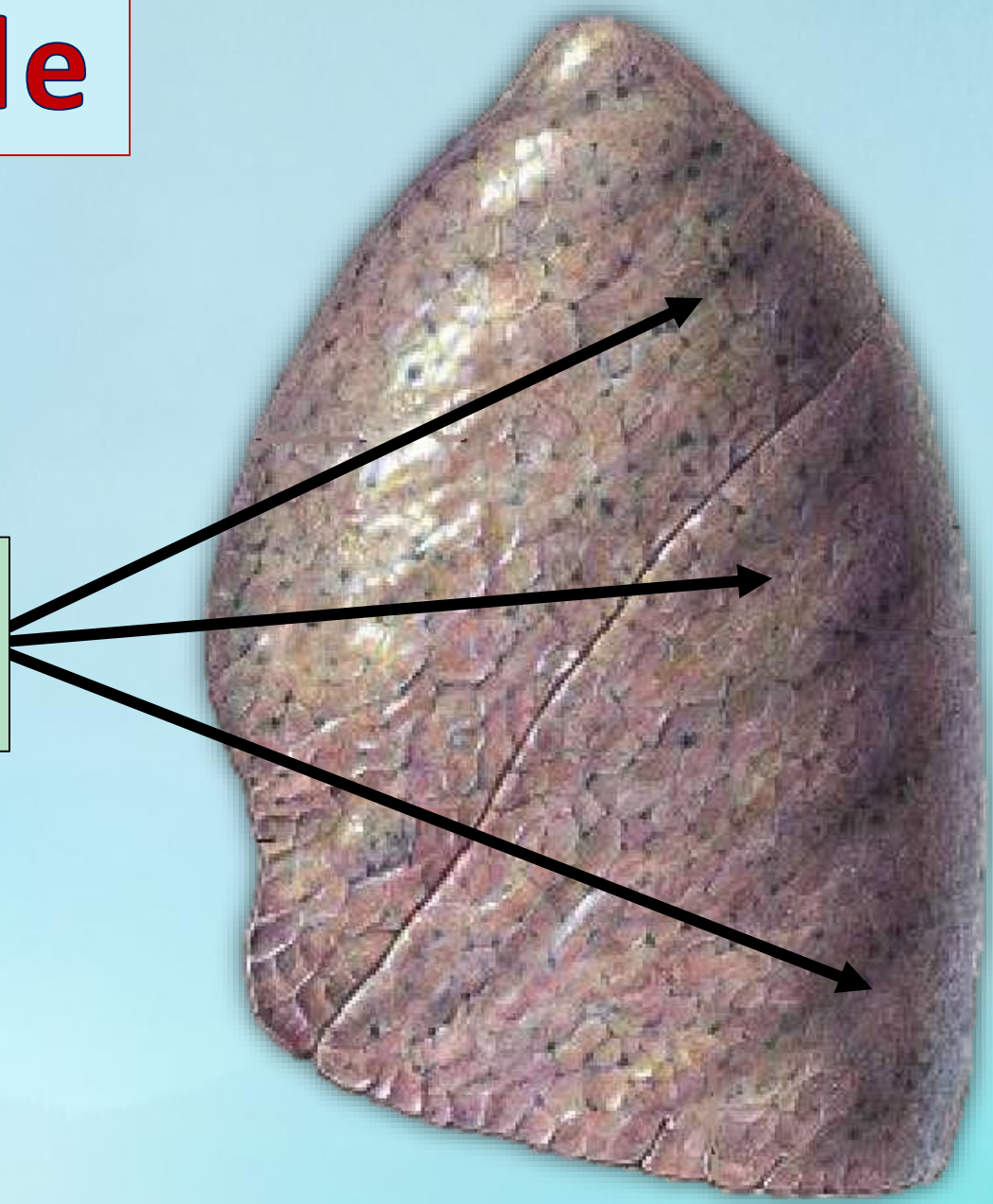




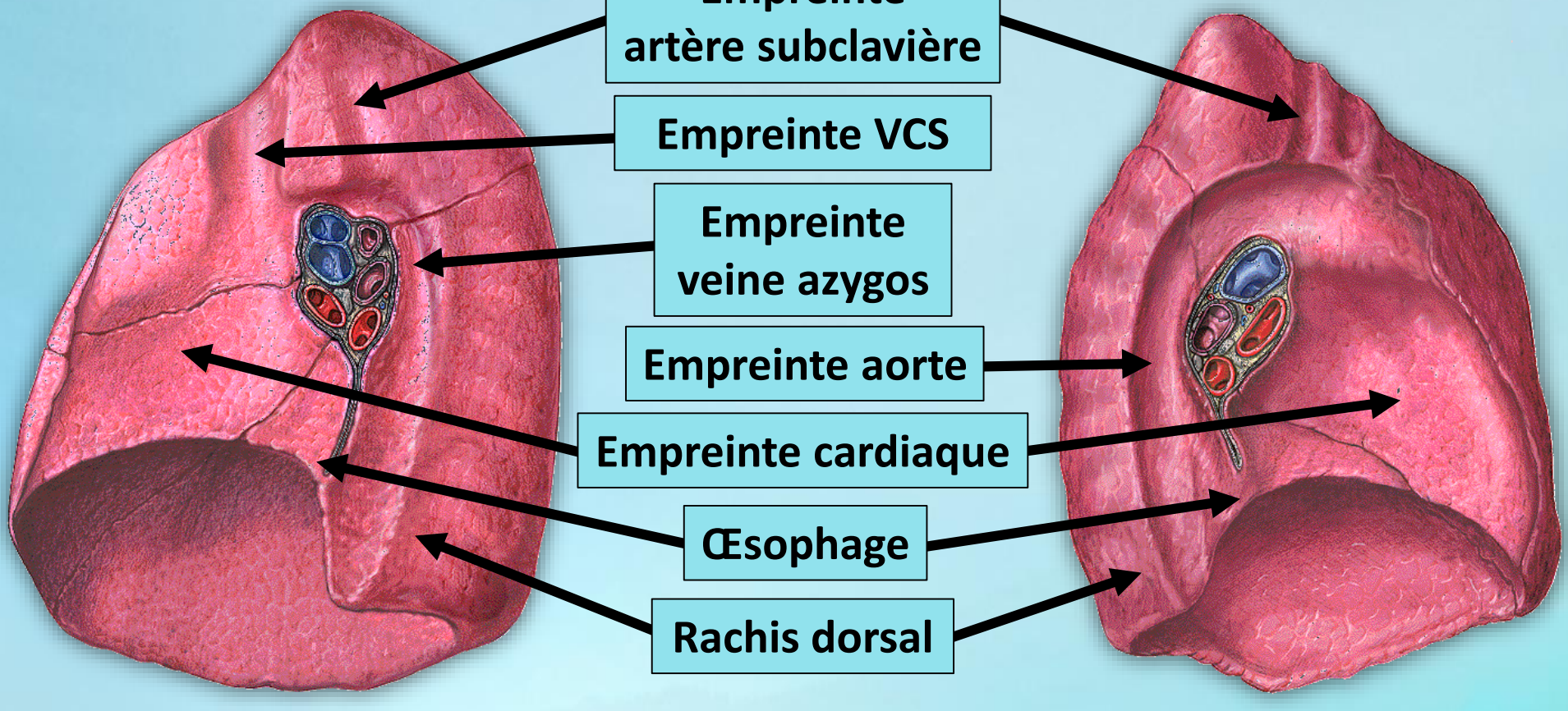

Rapports

Face latérale

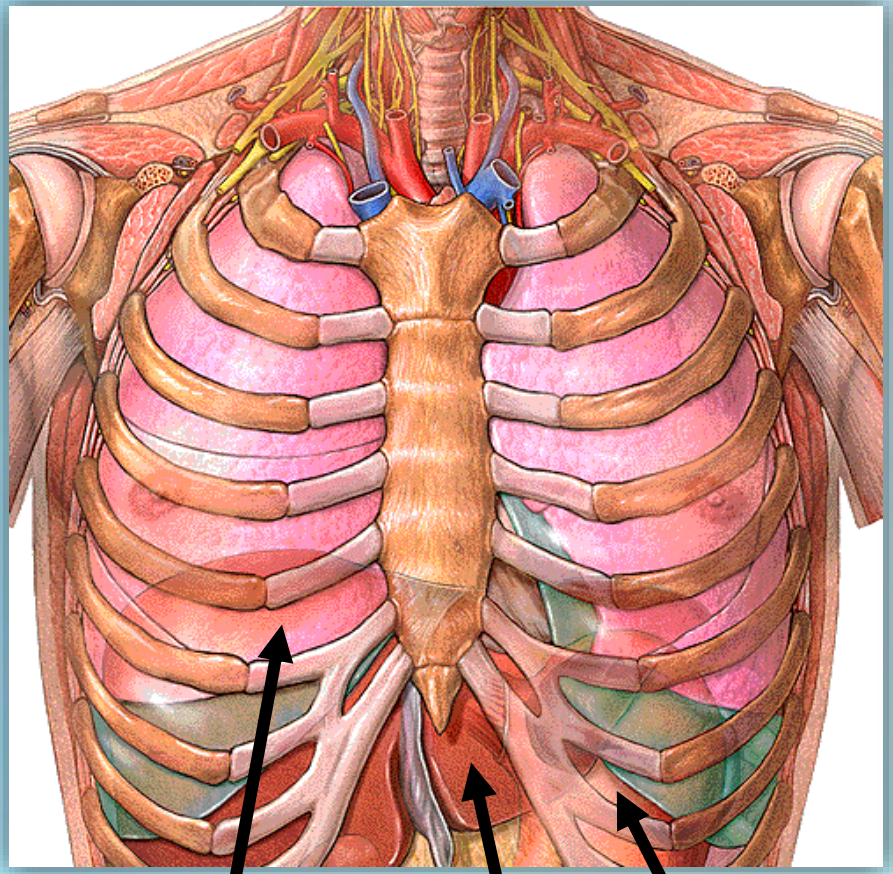
Marquée par:
empreintes costales

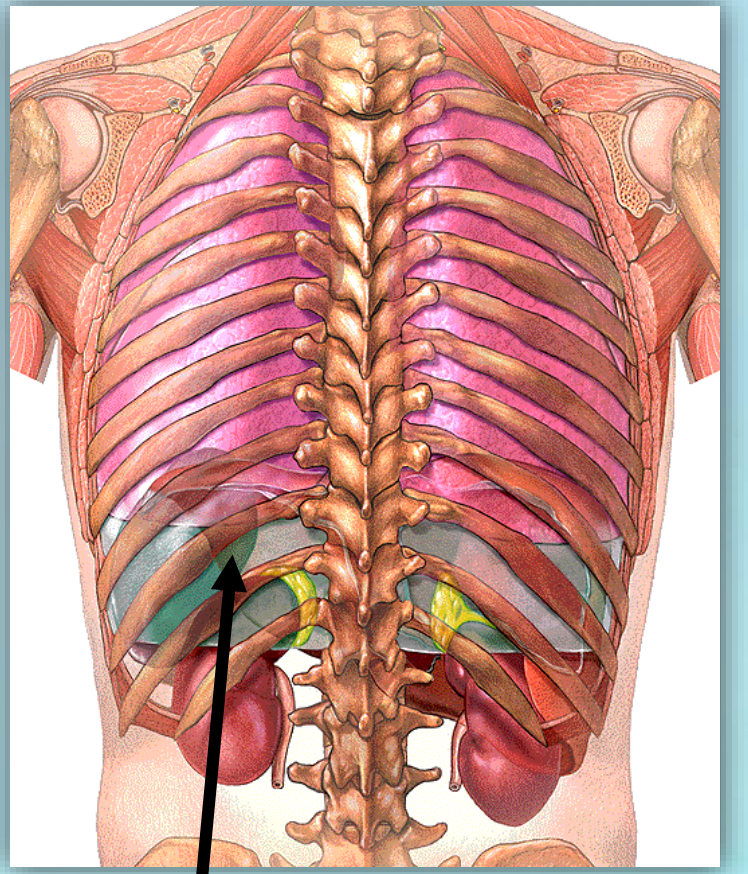


Face médiale



Base





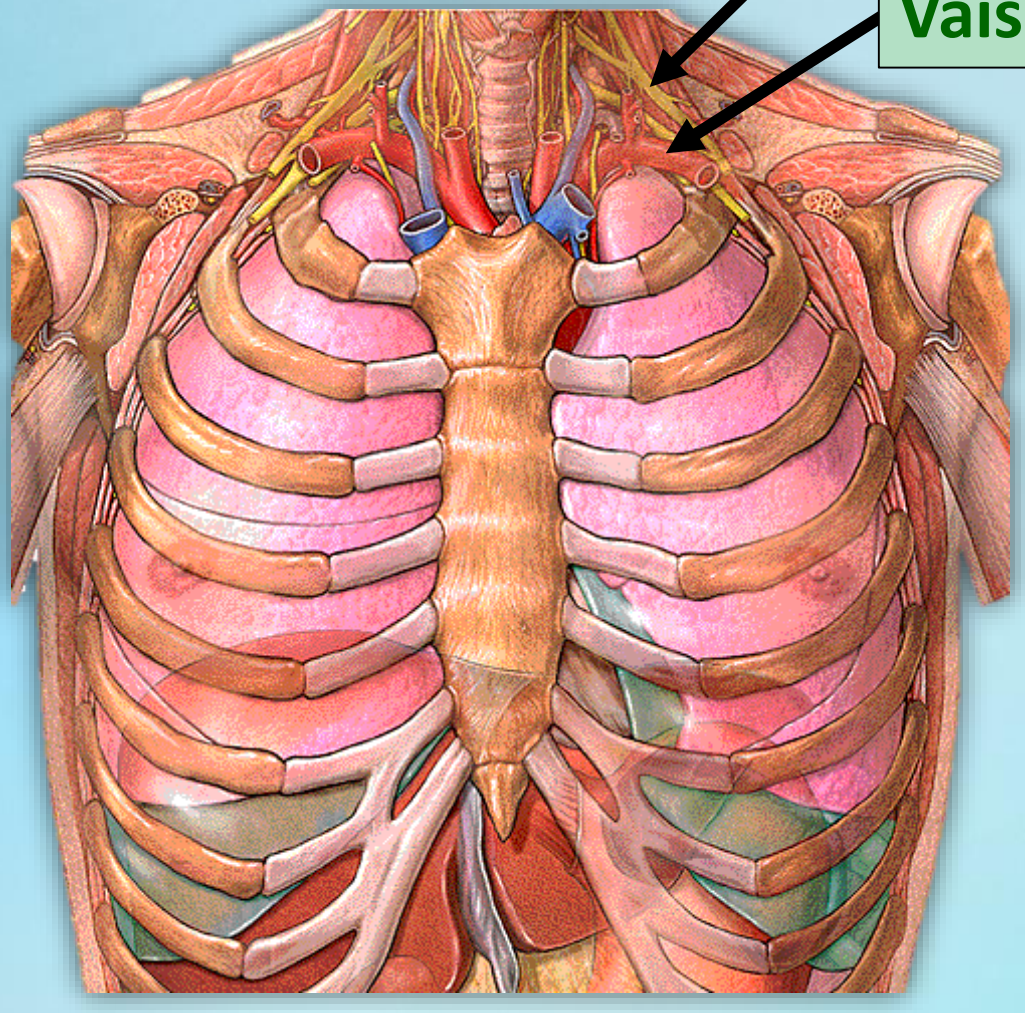
Diaphragme

Foie

Estomac
(à gauche)

Rate
(à gauche)

Apex




Bonne
respiration